Altlantis/Apollo Block Diagram

PCB STACK UP

- LAYER 1 : TOP
- LAYER 2 : SGND1
- LAYER 3 : IN1
- LAYER 4 : IN2
- LAYER 5 : VCC
- LAYER 6 : IN3
- LAYER 7 : SGND2
- LAYER 8 : BOT

RUN POWER SW PG 36,37

AC/BATT CONNECTOR PG 34

BATT CHARGER PG 34

CPU VR PG 38

DC/DC +3VSUS +5VSUS PG 33

Turion64 Sempron
25W/35W
AMD Socket S1 638P
PG 2,3,4

DDRII-SODIMM1 PG 5 — *DDRII 667mhz*

DDRII-SODIMM2 PG 5 — *DDRII 667mhz*

HT Link

nVIDIA C51M/D
Integraded VGA Function
468PIN
PG 6,7,8

PCI-E

Mini PCI-E Card
PCI Express Mini Card
(Wireless LAN/WAN) PG 27

Express Card (NEW CARD) PG 28

SIM CARD PG 27

PCI-Express 16X

NVDIA G3-64 for 15.4"
NVDIA G3-128 for 17"
Graphics PG 9,10,11,12,13,14

Option for 17" only
HDMI CON PG 18

TV_OUT
CRT_OUT
CRT/S-VIDEO PG 19

LVDS(2 Channel)
Panel Connector 15" / 17" PG 18

Cable Docking PG 31
- TV_OUT
- VGA
- RJ-45
- CIR/Pwr btn
- SPDIF Out
- Stereo MIC
- Headphone Jack
- USB Port
- VOL Cntr

HT Link

CLOCK INTERFACE

Option for 17" only

SATA - HDD PG 28 — *SATA2*

SATA - HDD PG 26 — *SATA0*

PATA- CD-ROM PG 26 — *PATA (66/100/133)*

nVIDIA MCP51M
508PIN
PG 15,16,17

USB2.0

Bluetooth PG 28

USB2.0 I/O Ports X3 PG 26

Camera 1.3MP PG 23

Mini PCI-E Card x1
Express Card x1
Cable Docking x1

PCI BUS / 33MHz

LAN

Azalia

REALTEK
RTL8201CL (10/100)/ REL8211B (10/100 /Gigabit)
PG 20

RICOH
RICOH 832
PG 21

RJ45 PG 20

IEEE1394 CONN PG 21

Memory CardReader PG 22

Conexant
Venice AMOM
CX20549-12
PG 23

Two-element microphone PG 23

Audio Jacks (Phone/SPDIF/ MIC) PG 23

LPC

ENE KBC
KB3920 Bx
PG 30

Keyboard Touch Pad PG 29

CIR PG 23

Capacitive Sense SW PG 29

X-Bus

FAN PG 31

Flash PG 30

AUDIO Amplifier PG 25

Jack to Speaker PG 25

MDC DAA PG 24

MODEM RJ 11 PG 20

VAULE DEFINE
A=0603,B=0805,C=1206,F=1%,
OTHER IS 0402

EXAMPLE
10R=10ohm(0402)
10A=10ohm(0603)
10B=10ohm(0805)
10C=10ohm(1206)
10/F=10ohm(0402 and 1%)

PCI ROUTING TABLE	IDSEL	INTERUPT	DEVICE
REQ0# / GNT0#	AD21	INTA#,INTB#	RICOH832

PROJECT : AT8
Quanta Computer Inc.

NB5/RD2/HW1

Size: Custom | Document Number: Block Diagram | Rev: 1A

Date: Thursday, December 29, 2005 | Sheet 1 of 38

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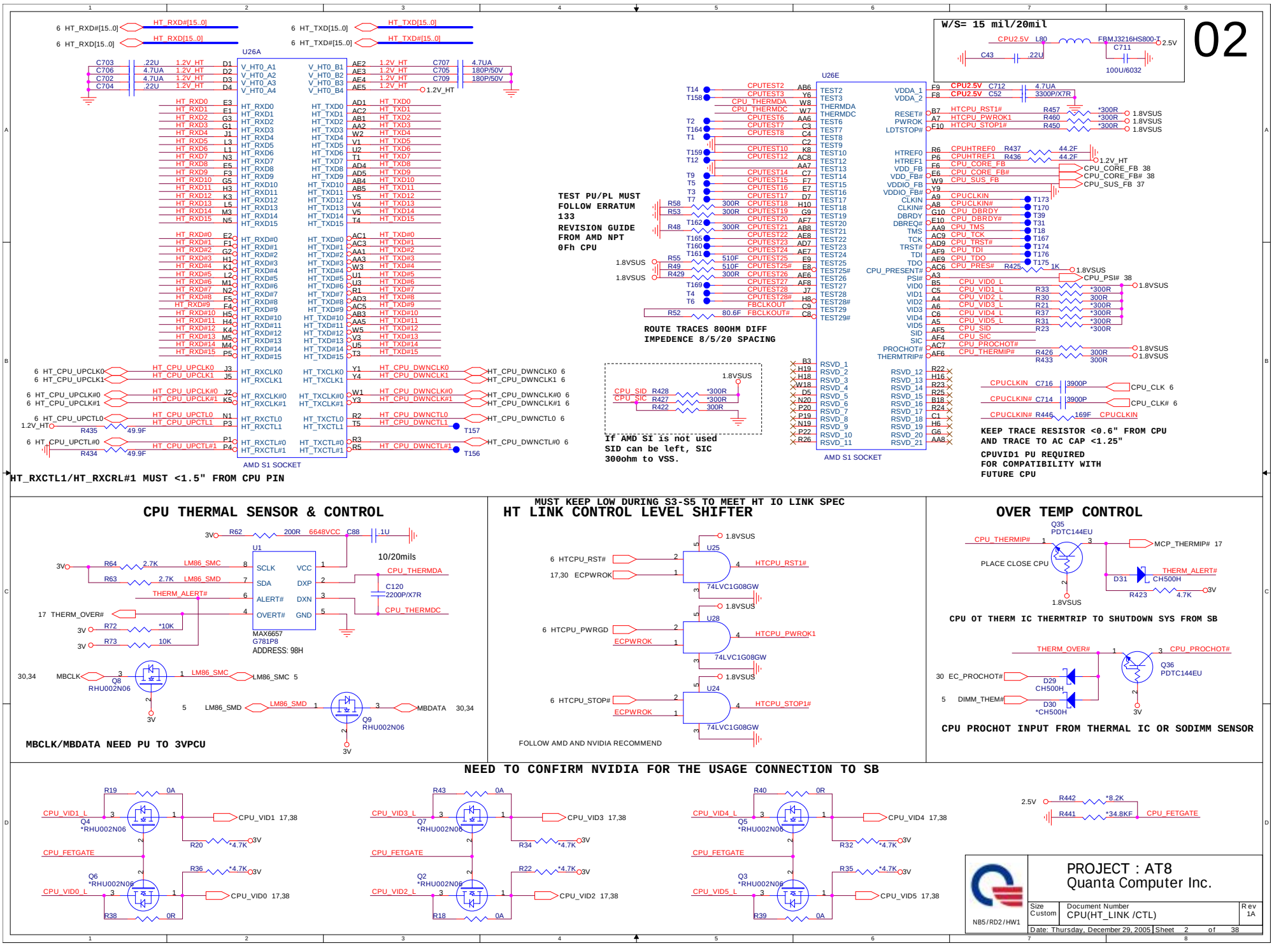

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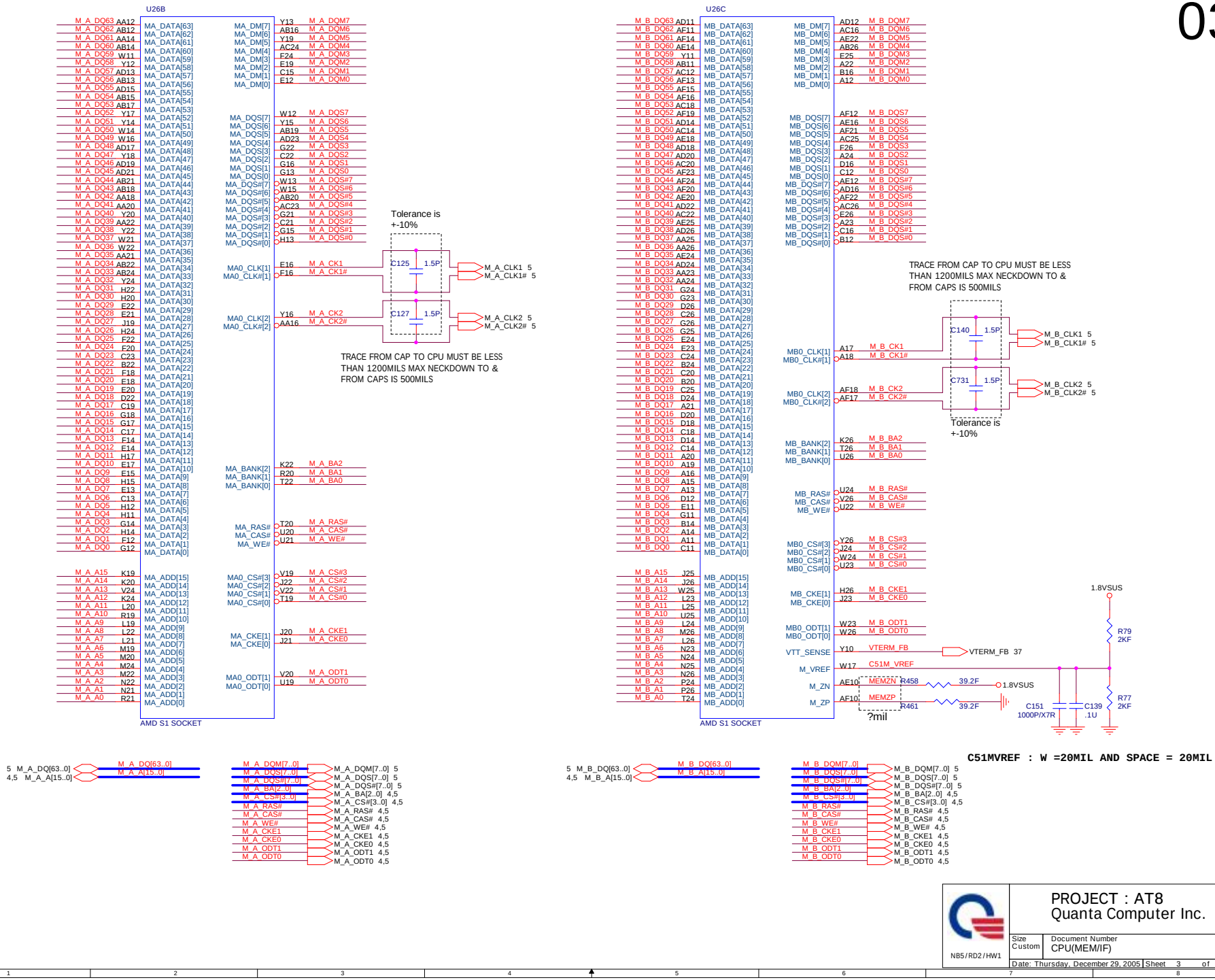

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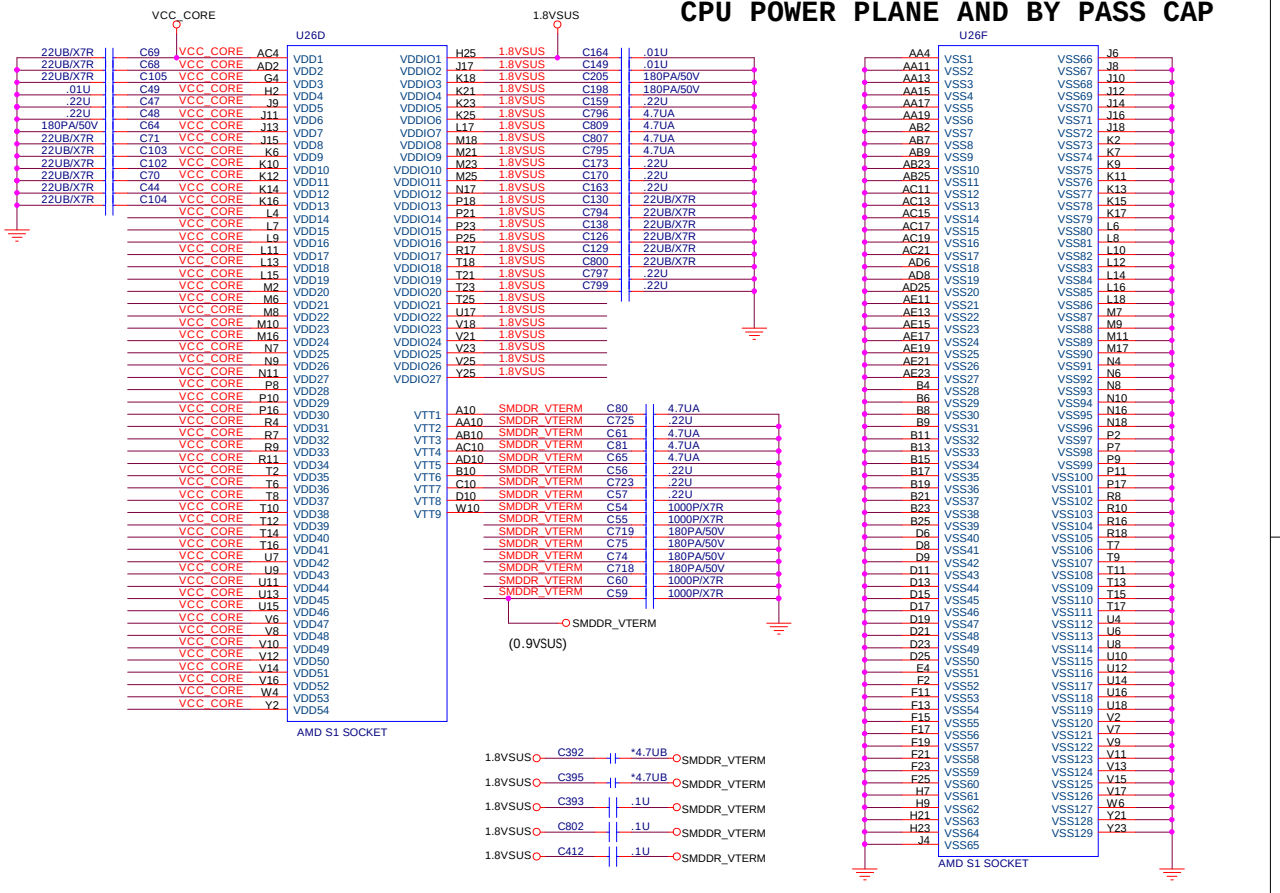
CPU POWER PLANE AND BY PASS CAP

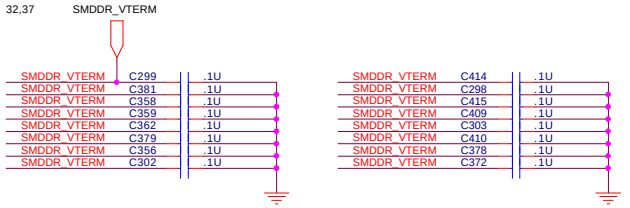


DDR2 TERMINATION BYPASS CAP

04

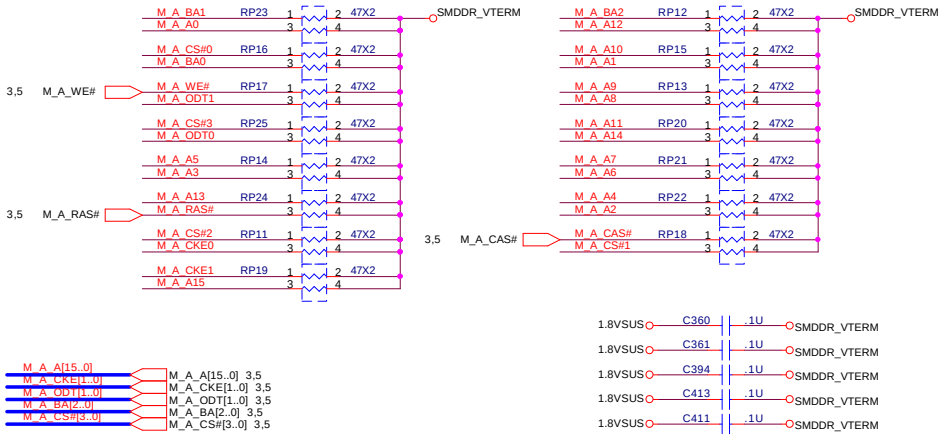
CHANNEL A

CHANNEL B

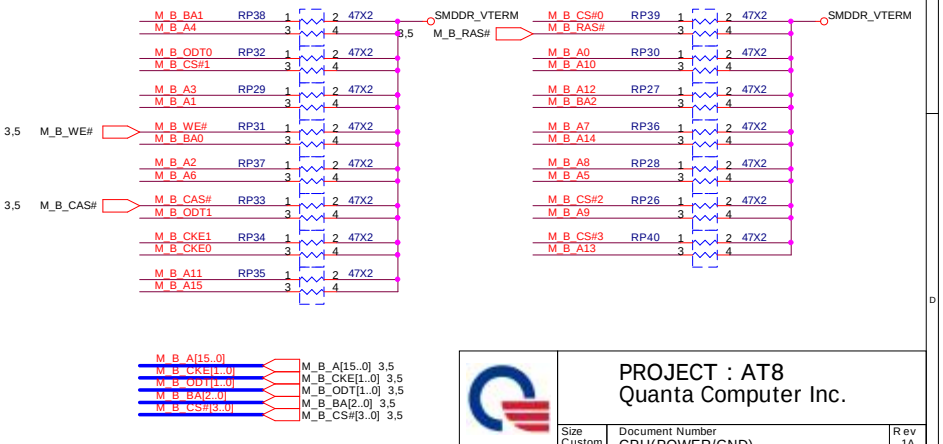


Layout note: Place one cap close to every 2 pullup resistors terminated to SMDDR_VTERM

DDRII CHANNEL A TERMINATION



DDRII CHANNEL B TERMINATION



NB5/RD2/HW1

PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	CPU(POWER/GND)	1A

Date: Thursday, December 29, 2005 Sheet 4 of 38

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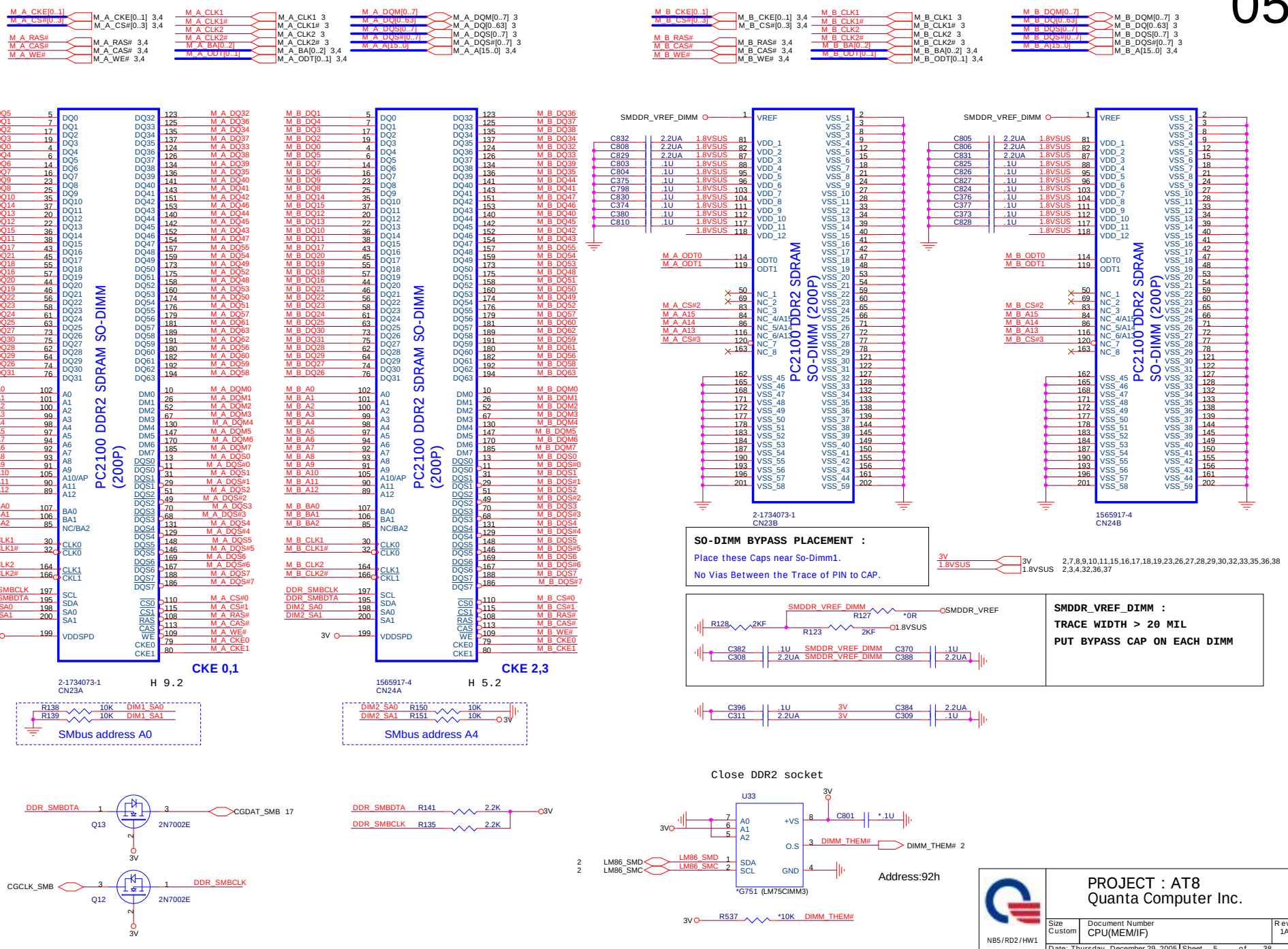

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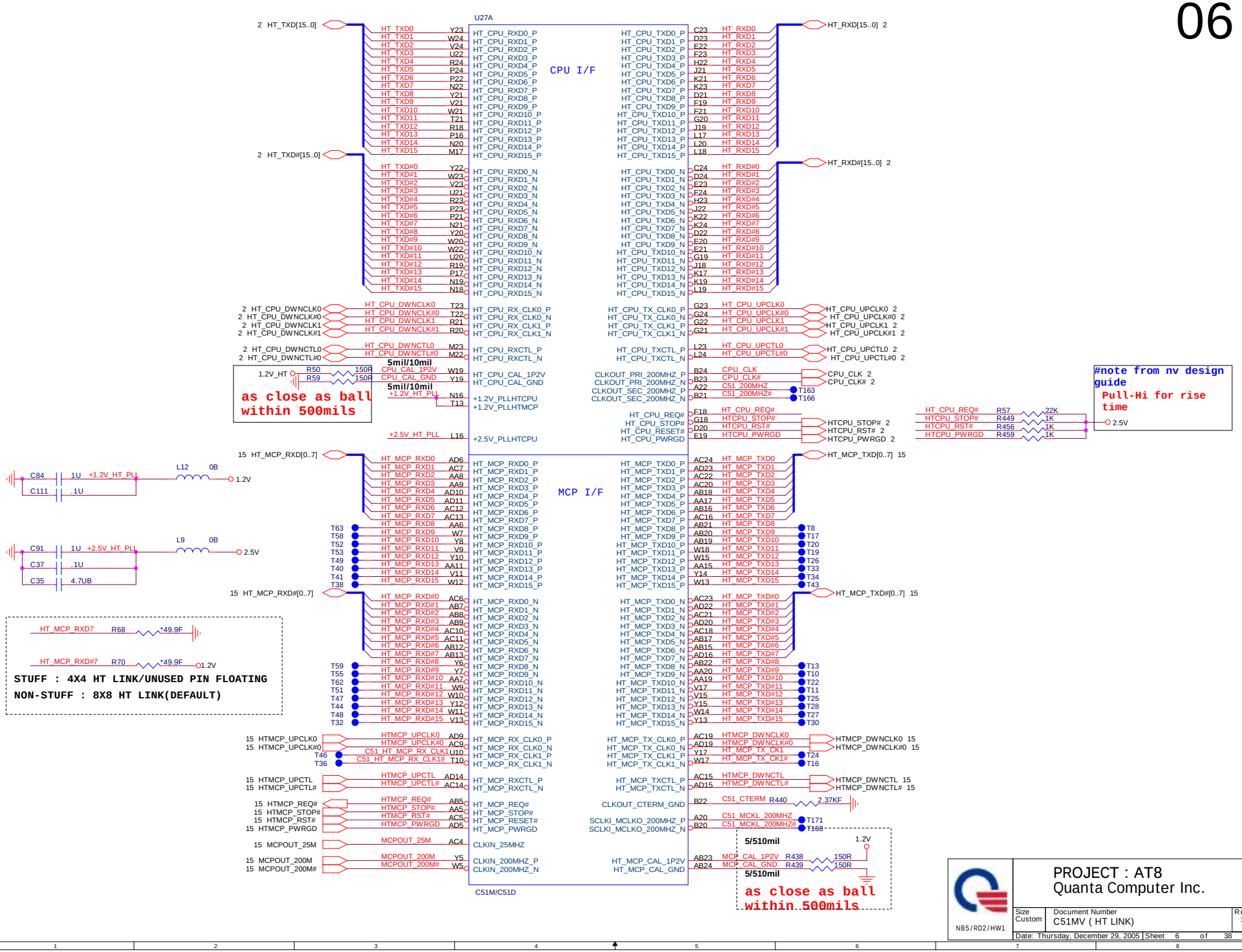

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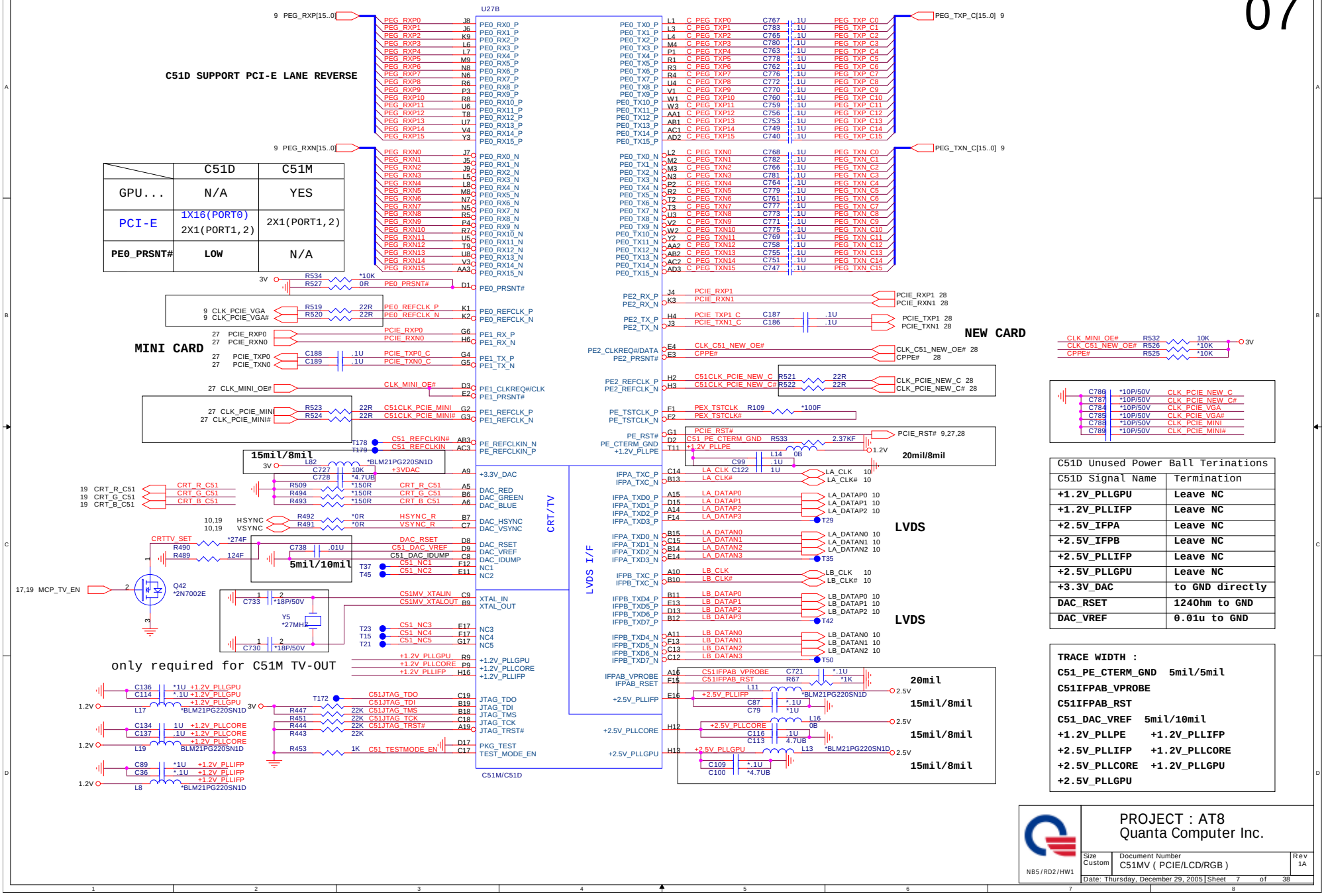

	C51D	C51M
GPU...	N/A	YES
PCI-E	1X16(PORT0) 2X1(PORT1,2)	2X1(PORT1,2)
PE0_PRSNT#	LOW	N/A

C51D Unused Power Ball Terinations	
C51D Signal Name	Termination
+1.2V_PLLGPU	Leave NC
+1.2V_PLLIFP	Leave NC
+2.5V_IFPA	Leave NC
+2.5V_IFPB	Leave NC
+2.5V_PLLIFP	Leave NC
+2.5V_PLLGPU	Leave NC
+3.3V_DAC	to GND directly
DAC_RSET	124Ohm to GND
DAC_VREF	0.01u to GND

TRACE WIDTH :

C51_PE_CTERM_GND 5mil/5mil
C51IFPAB_VPROBE
C51IFPAB_RST
C51_DAC_VREF 5mil/10mil
+1.2V_PLLPE +1.2V_PLLIFP
+2.5V_PLLIFP +1.2V_PLLCORE
+2.5V_PLLCORE +1.2V_PLLGPU
+2.5V_PLLGPU

PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	C51MV (PCIE/LCD/RGB)	1A

Date: Thursday, December 29, 2005 | Sheet 7 of 38

NB5/RD2/HW1

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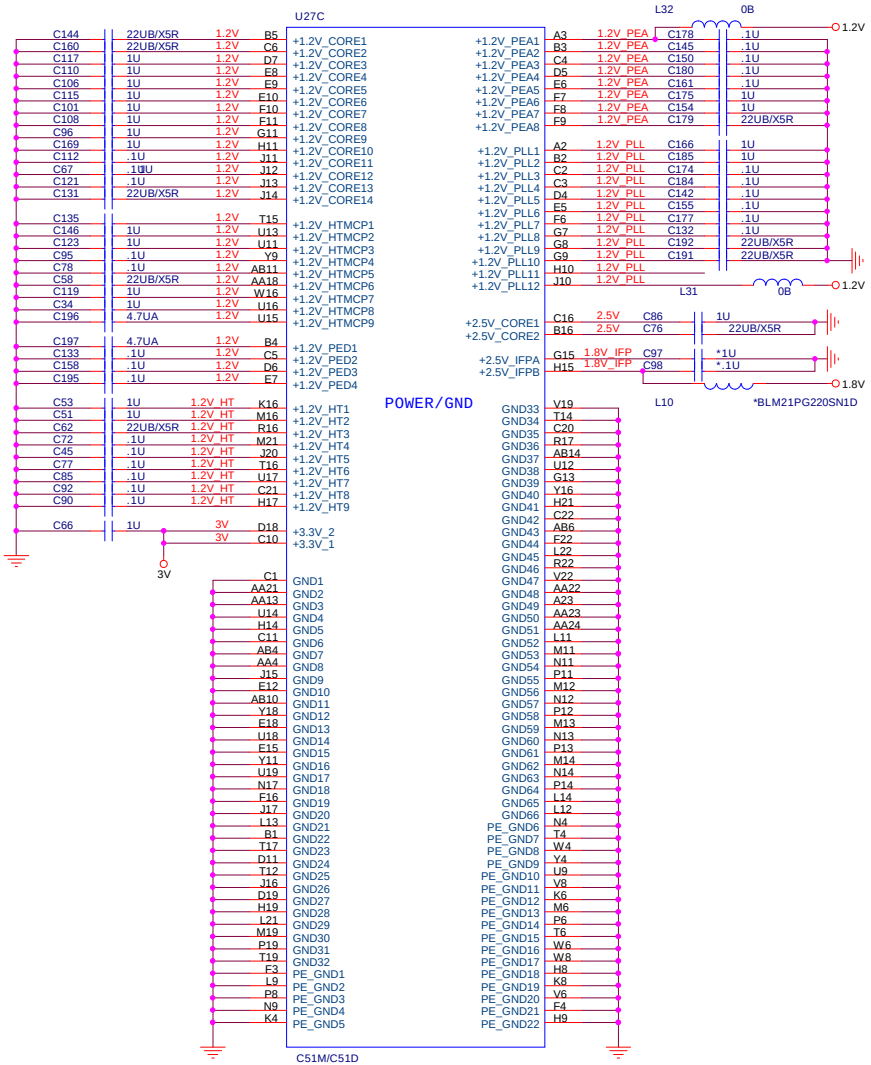
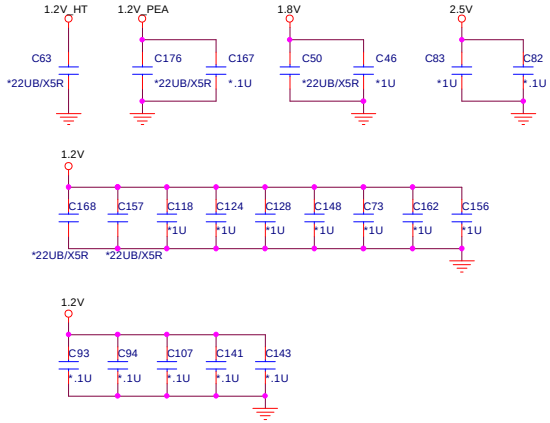
C51M/C51D POWER PLANE/GND & BYPASS





PROJECT : AT8 Quanta Computer Inc.		
NB5/RD2/HW1	Size: Custom Document Number: C51MV (POWER/GND)	Rev: 1A
	Date: Wednesday, January 11, 2006 Sheet 8 of 38	

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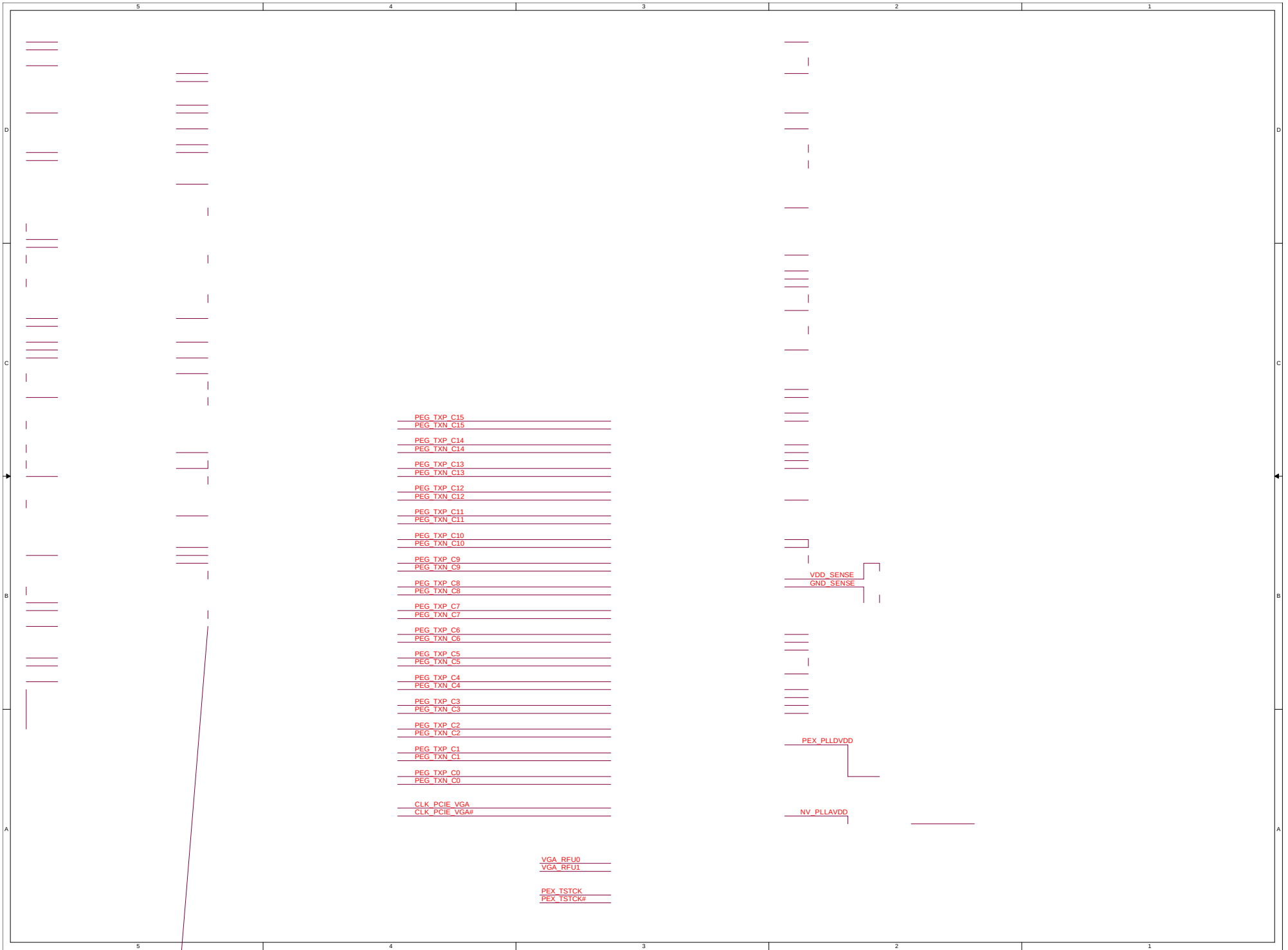
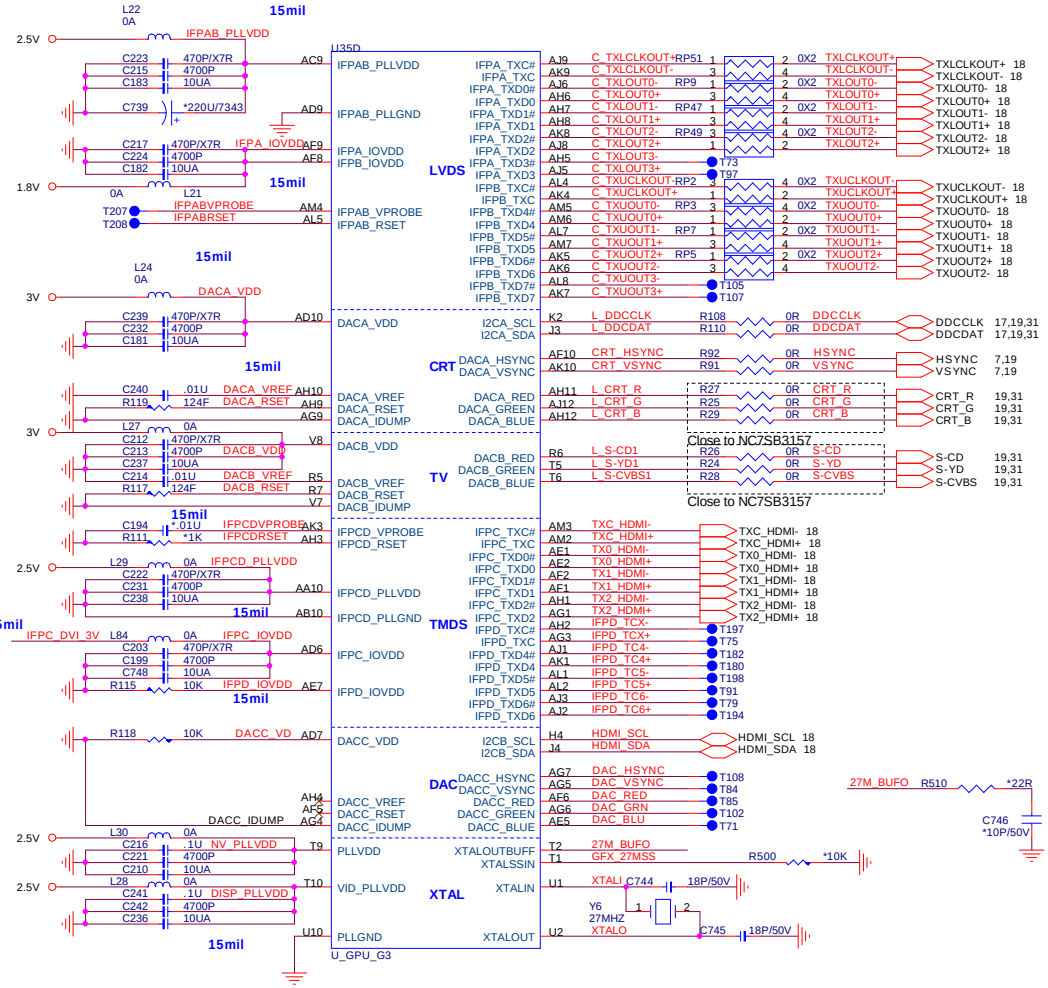
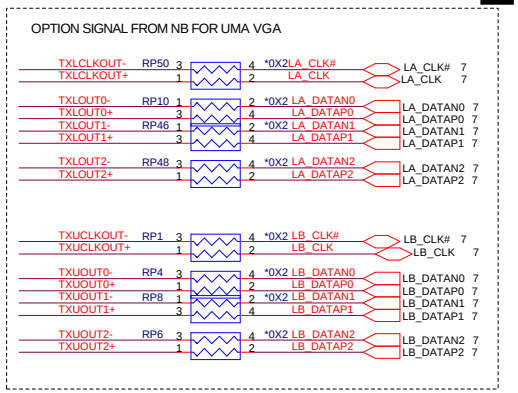
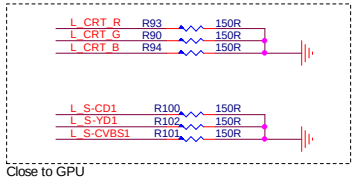
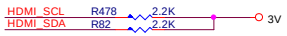
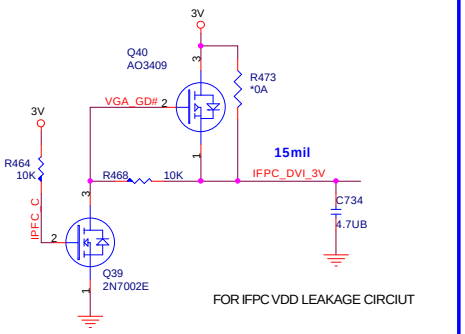

PEG_TXP_C15
PEG_TXN_C15
PEG_TXP_C14
PEG_TXN_C14
PEG_TXP_C13
PEG_TXN_C13
PEG_TXP_C12
PEG_TXN_C12
PEG_TXP_C11
PEG_TXN_C11
PEG_TXP_C10
PEG_TXN_C10
PEG_TXP_C9
PEG_TXN_C9
PEG_TXP_C8
PEG_TXN_C8
PEG_TXP_C7
PEG_TXN_C7
PEG_TXP_C6
PEG_TXN_C6
PEG_TXP_C5
PEG_TXN_C5
PEG_TXP_C4
PEG_TXN_C4
PEG_TXP_C3
PEG_TXN_C3
PEG_TXP_C2
PEG_TXN_C2
PEG_TXP_C1
PEG_TXN_C1
PEG_TXP_C0
PEG_TXN_C0
CLK_PCIE_VGA
CLK_PCIE_VGA#
VGA_RFU0
VGA_RFU1
PEX_TSTCK
PEX_TSTCK#
VDD_SENSE
GND_SENSE
PEX_PLLDVDD
NV_PLLAVDD

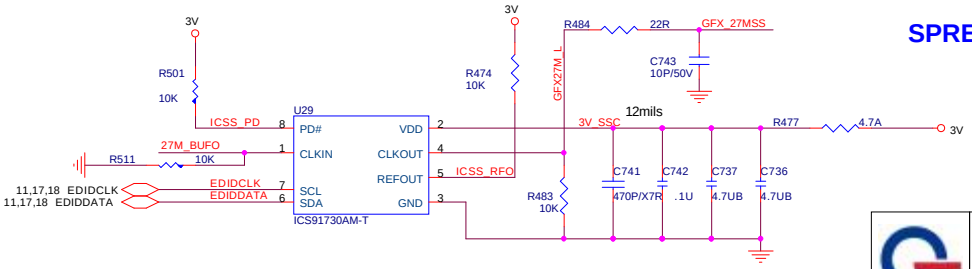












SPREAD SPECTRUM

I2C ADDRESS: 0xD4H

PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	GFX(LVDS, CRT, TV)	1A

NB5/RD2/HW1

Date: Thursday, December 29, 2005 | Sheet 10 of 38

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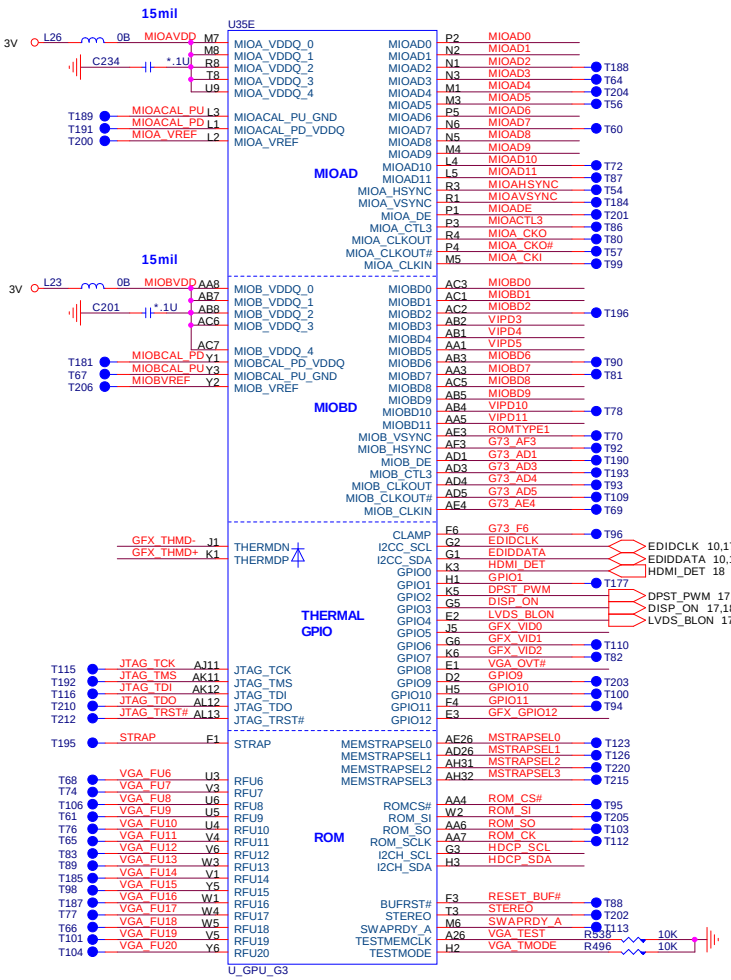
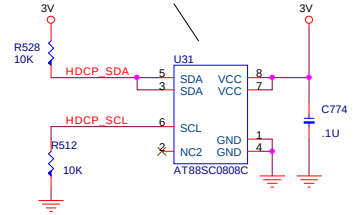
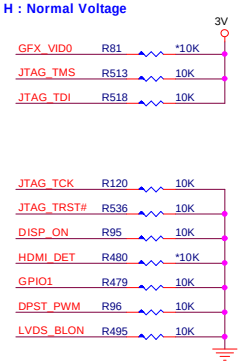


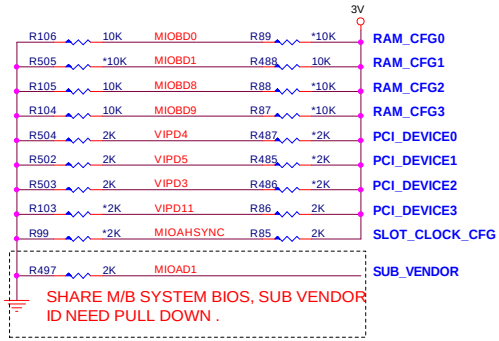


PCI DEVICE

PCI_DEVICE[3:0]	DESCRIPTION
1000	G72M/G73M
0111	G72M-V/G73M-V
others	Reserved

GFX_VID0
L : Low Voltage
H : Normal Voltage



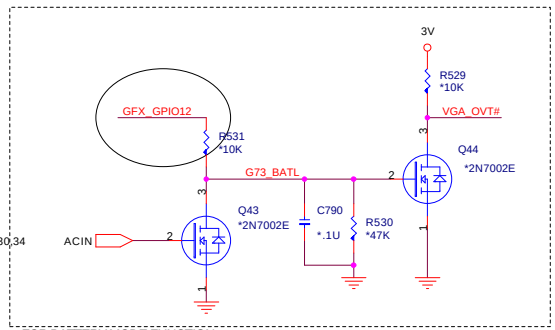


G72M VRAM Configuration Table

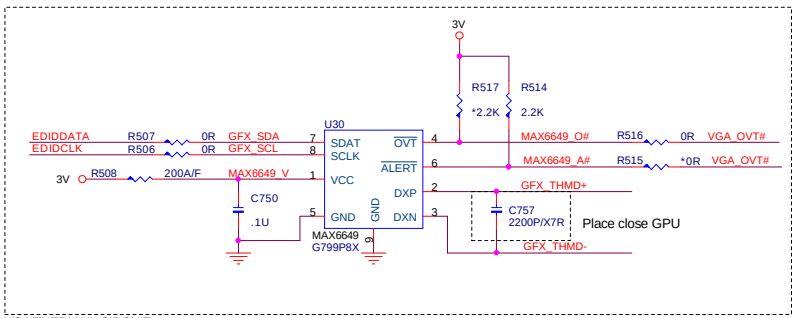
RAM_CFG[3:0]	DESCRIPTION	Vendor
0000	DDR2 16Mx16x4, 64bit, 128MB	Elpida
0001	DDR2 16Mx16x4, 64bit, 128MB	Samsung
0010	DDR2 16Mx16x4, 64bit, 128MB	Infineon
0011	DDR2 16Mx16x4, 64bit, 128MB	Hynix
0100	Reserved	
0101	DDR2 32Mx16x4, 64bit, 256MB	Samsung
0110	DDR2 32Mx16x4, 64bit, 256MB	Infineon
0111	DDR2 32Mx16x4, 64bit, 256MB	Hynix
1000	DDR2 16Mx16x2, 32bit, 64MB	Elpida
1001	DDR2 16Mx16x2, 32bit, 64MB	Samsung
1010	DDR2 16Mx16x2, 32bit, 64MB	Infineon
1011	DDR2 16Mx16x2, 32bit, 64MB	Hynix
others	Reserved	

G73M VRAM Configuration Table

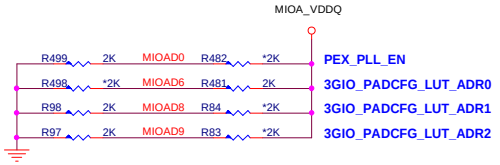
RAM_CFG[3:0]	DESCRIPTION	Vendor
0000	DDR2 16Mx16x8, 128bit, 256MB	Elpida
0001	DDR2 16Mx16x8, 128bit, 256MB	Samsung
0010	DDR2 16Mx16x8, 128bit, 256MB	Infineon
0011	DDR2 16Mx16x8, 128bit, 256MB	Hynix
0100	Reserved	
0101	DDR2 32Mx16x8, 128bit, 512MB	Samsung
0110	DDR2 32Mx16x8, 128bit, 512MB	Infineon
0111	DDR2 32Mx16x8, 128bit, 512MB	Hynix
1000	DDR2 16Mx16x4, 64bit, 128MB	Elpida
1001	DDR2 16Mx16x4, 64bit, 128MB	Samsung
1010	DDR2 16Mx16x4, 64bit, 128MB	Infineon
1011	DDR2 16Mx16x4, 64bit, 128MB	Hynix
1100	Reserved	
1101	DDR2 32Mx16x4, 64bit, 256MB	Samsung
1110	DDR2 32Mx16x4, 64bit, 256MB	Infineon
1111	DDR2 32Mx16x4, 64bit, 256MB	Hynix



FOR BATTERY MODE FUNCTION
GFX_GPIO12=H ENABLE
GFX_GPIO12=L DISABLE



VGA THERMAIL CIRCUIT



PROJECT : AT8
Quanta Computer Inc.

NB5/RD2/HW1

Size	Document Number	Rev
Custom	GFX(ROM, GPIO, STRAP)	1A

Date: Thursday, December 29, 2005 Sheet 11 of 38

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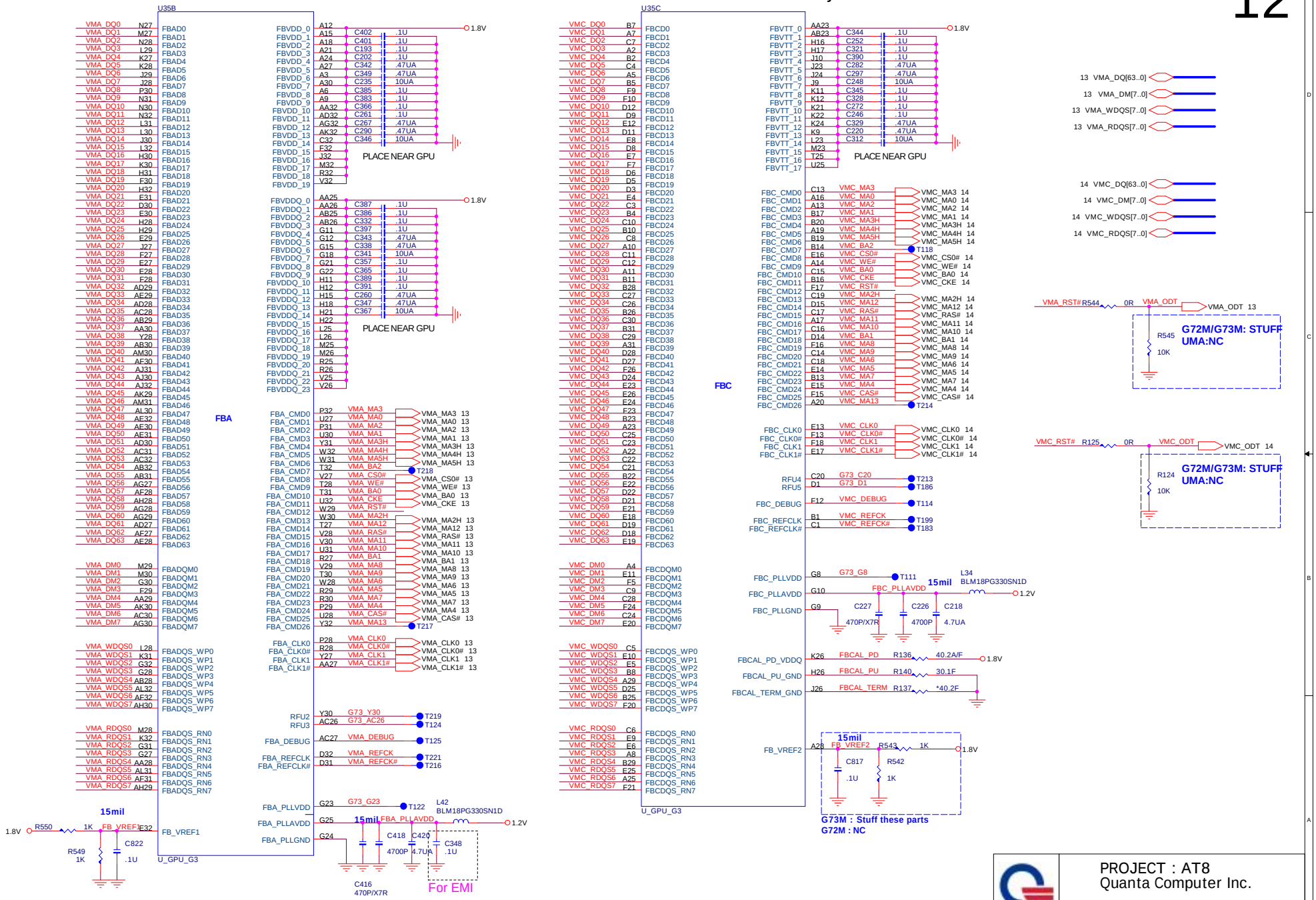
Channel C is available on G73M only



VREF = FBVDDQ * Rbot/(Rtop + Rbot)

NB5/RD2/HW1	PROJECT : AT8 Quanta Computer Inc.	
	Size: Custom Document Number: GFX (MEM INTERFACE)	Rev: 1A
	Date: Thursday, December 29, 2005 Sheet 12 of 38	

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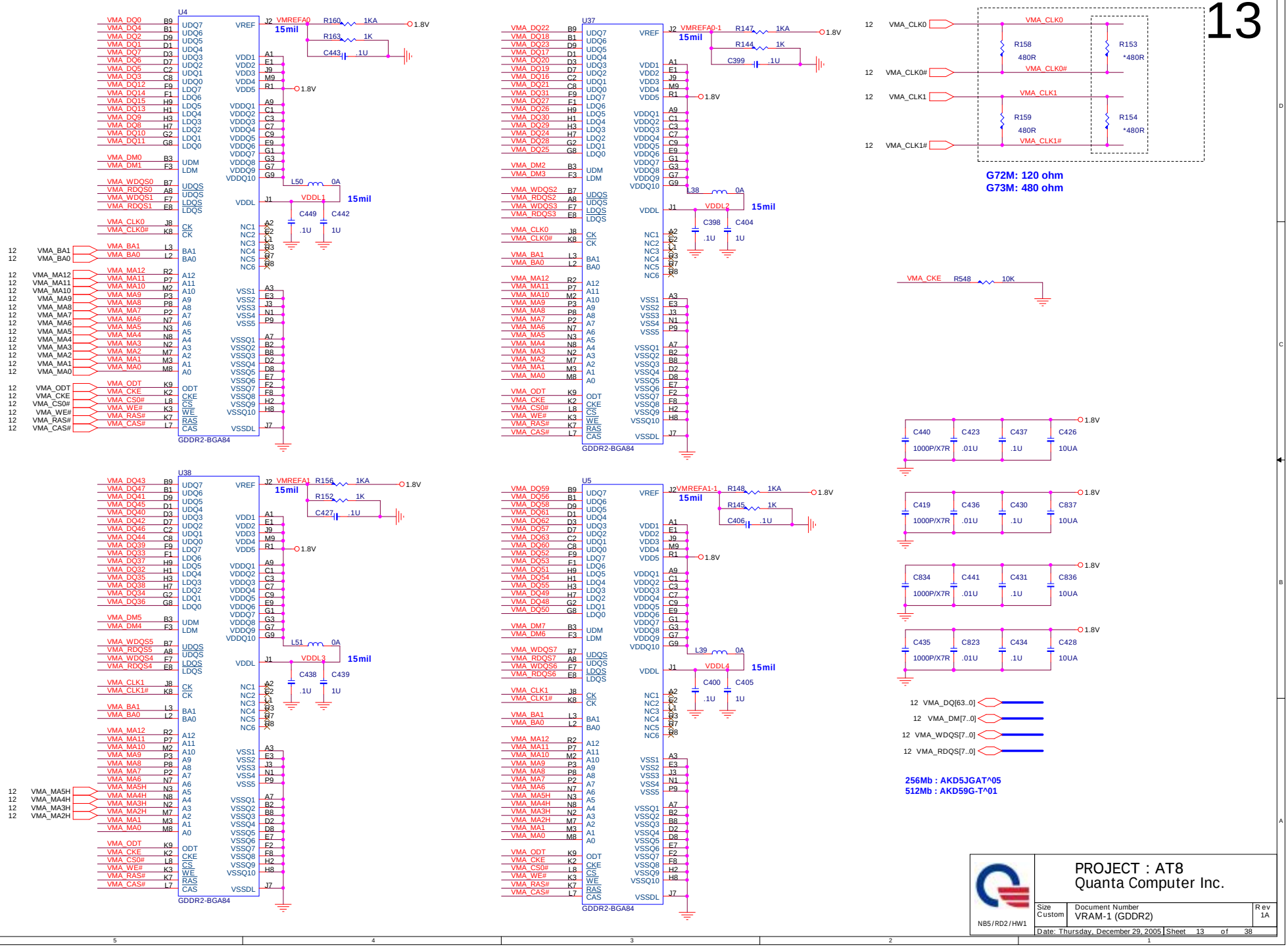

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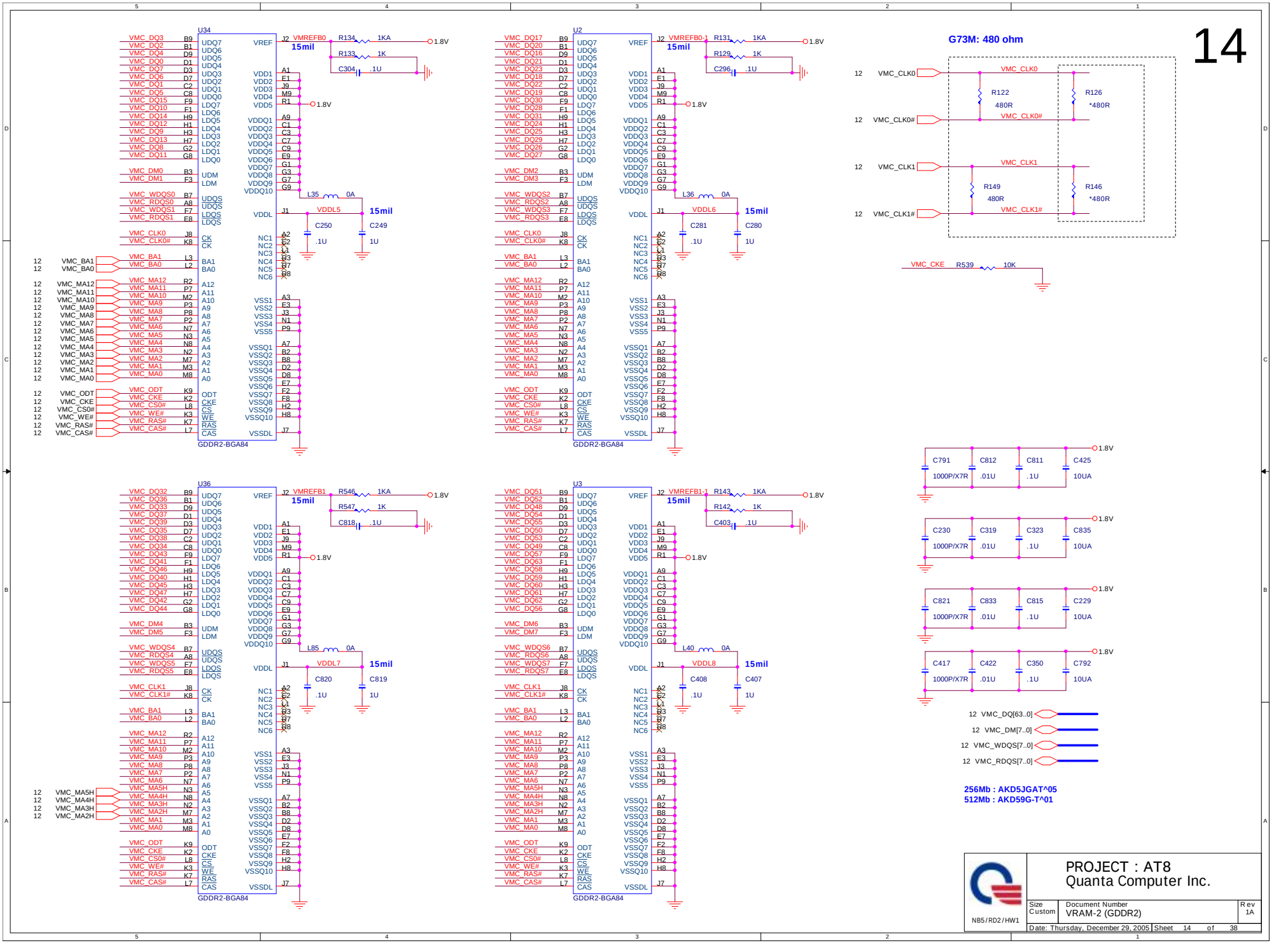

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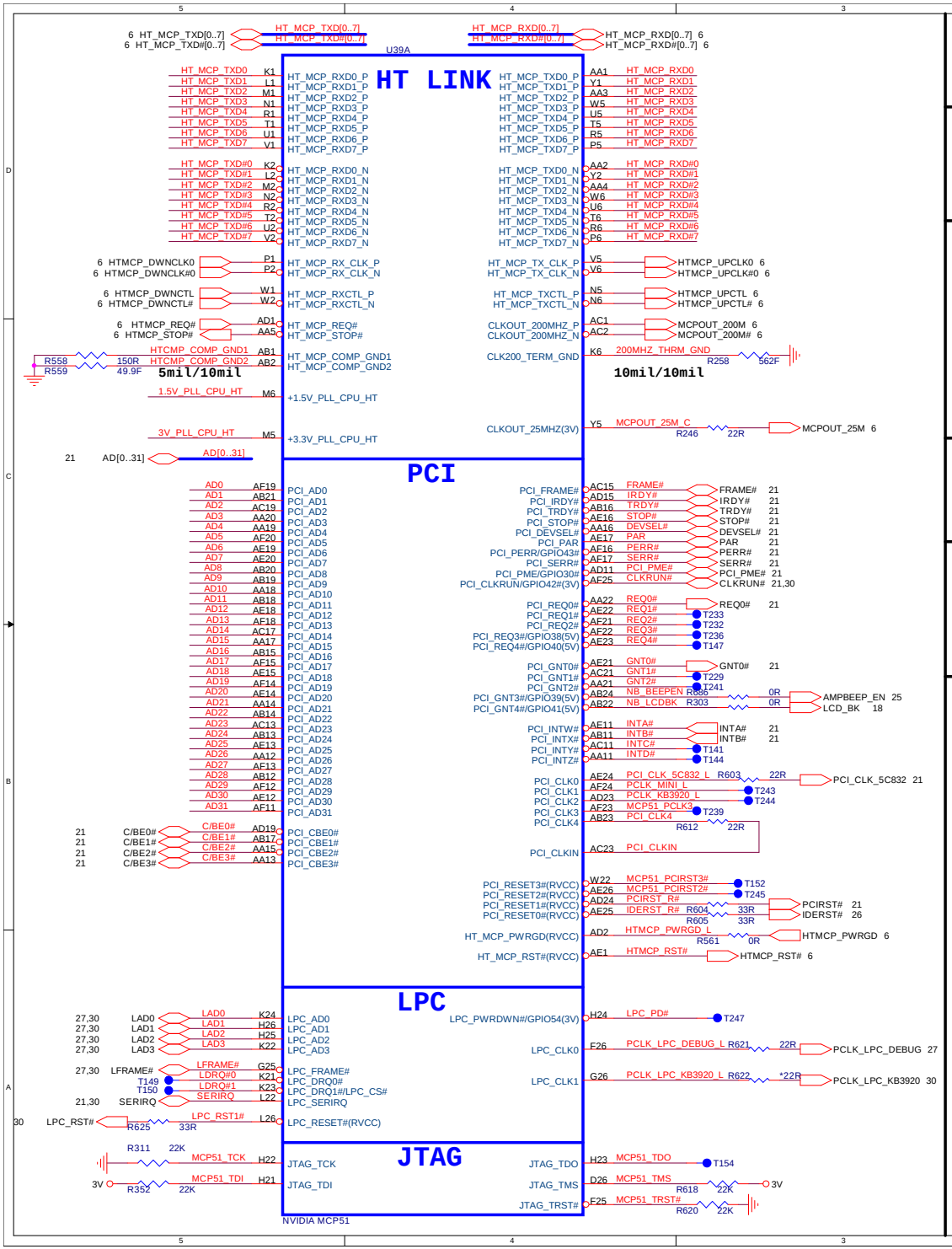

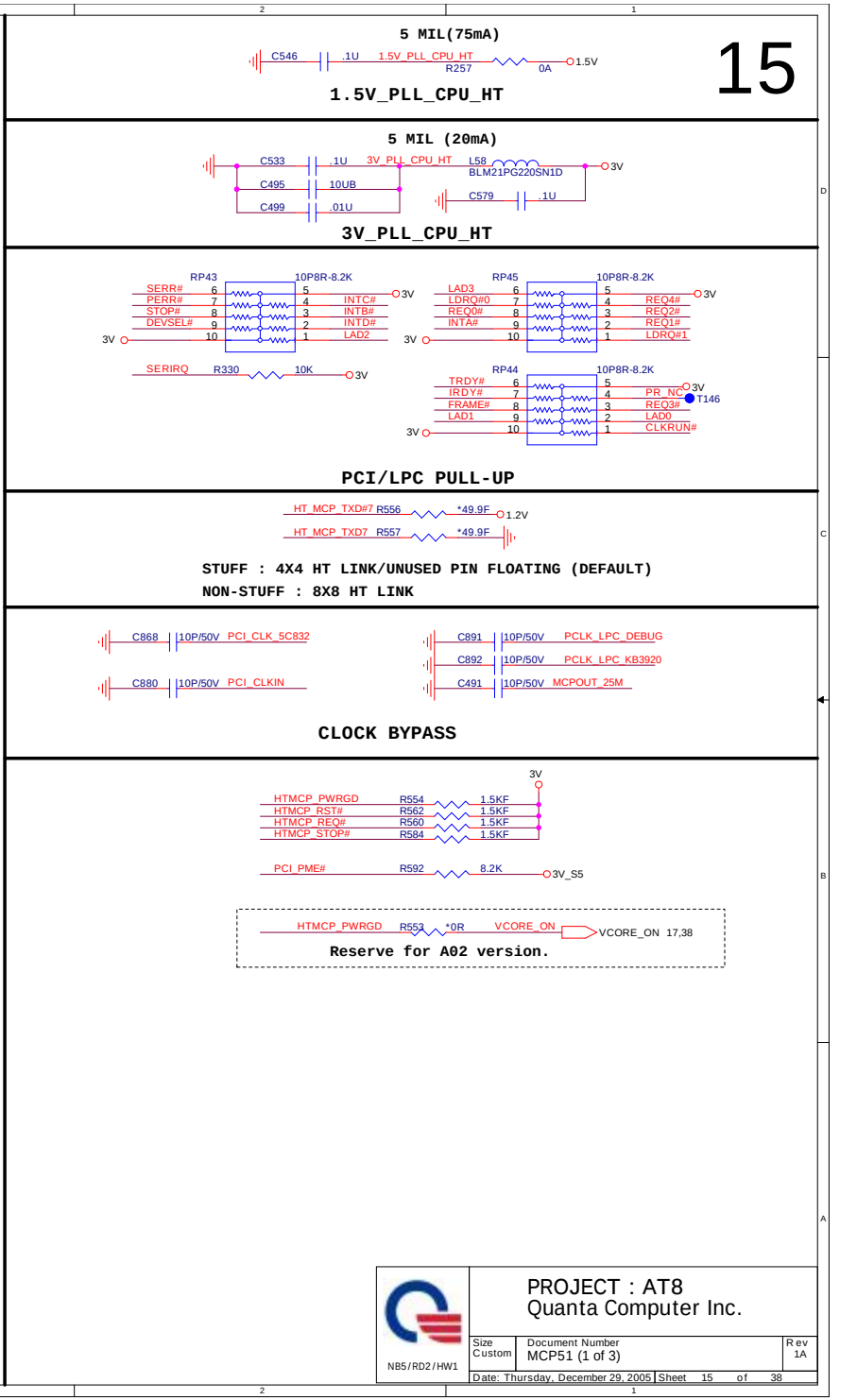

15

PROJECT : AT8
Quanta Computer Inc.

NB5/RD2/HW1

Size	Document Number	Rev
Custom	MCP51 (1 of 3)	1A

Date: Thursday, December 29, 2005 Sheet 15 of 38

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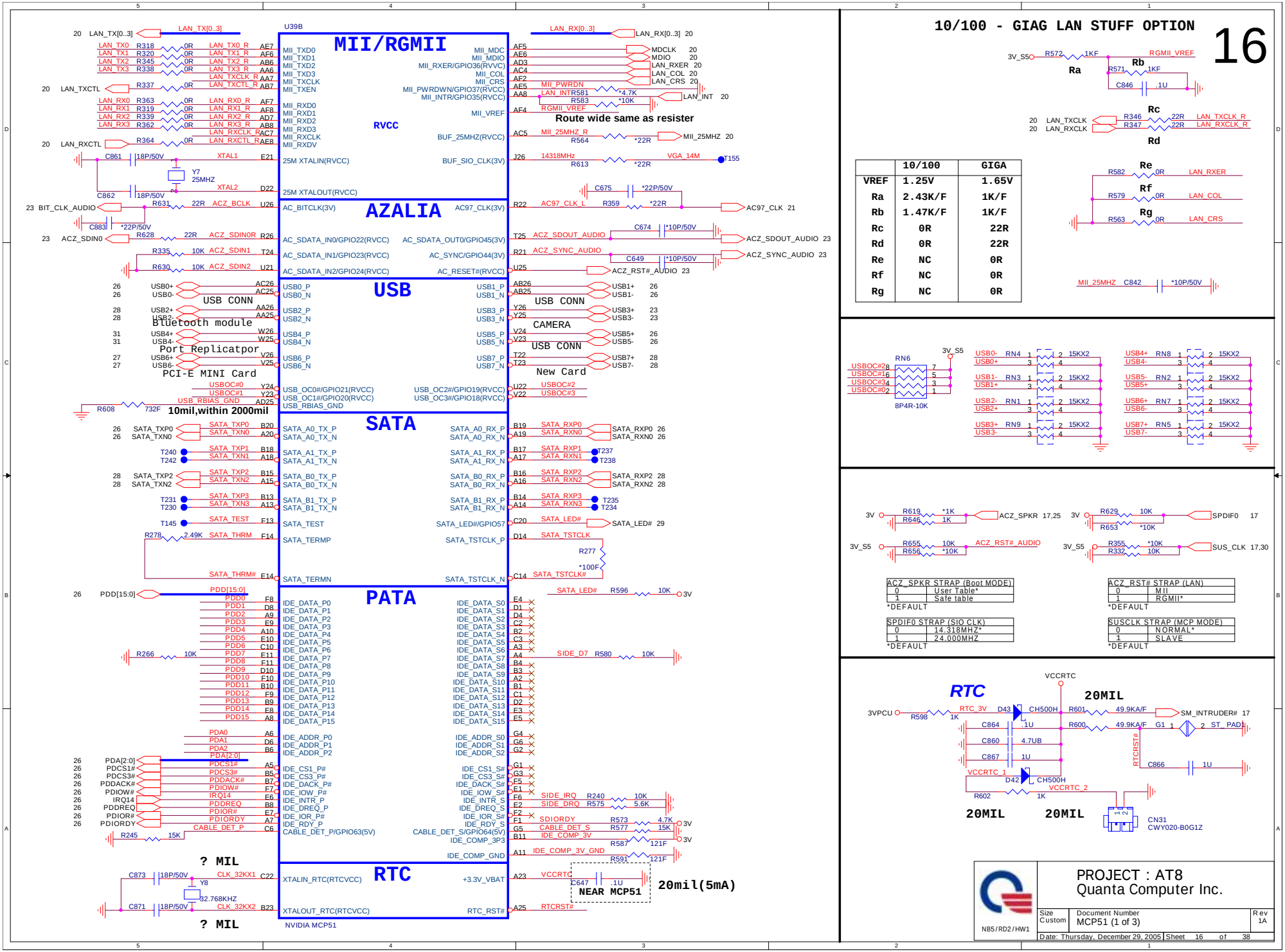

	10/100	GIGA
VREF	1.25V	1.65V
Ra	2.43K/F	1K/F
Rb	1.47K/F	1K/F
Rc	0R	22R
Rd	0R	22R
Re	NC	0R
Rf	NC	0R
Rg	NC	0R

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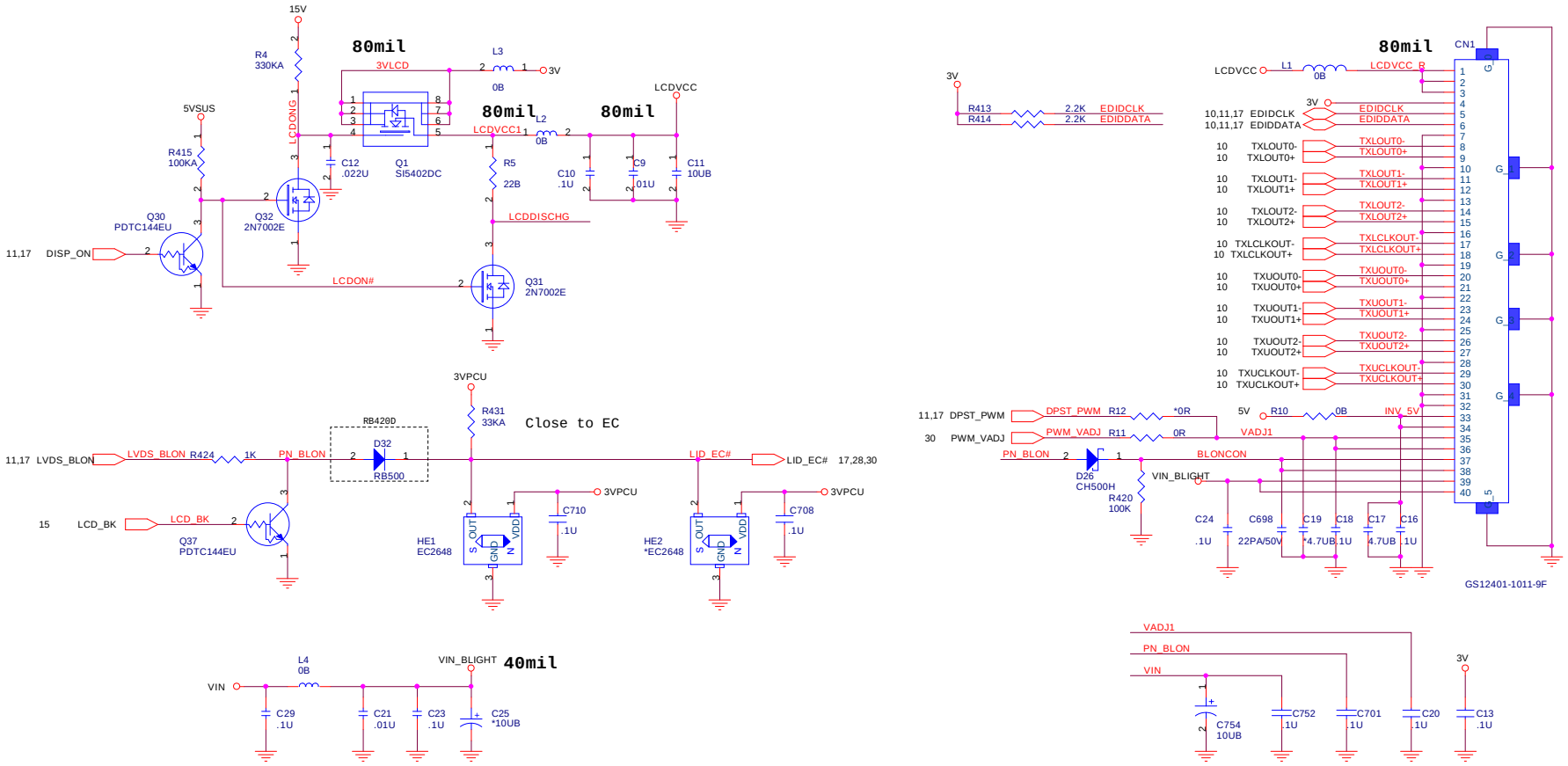
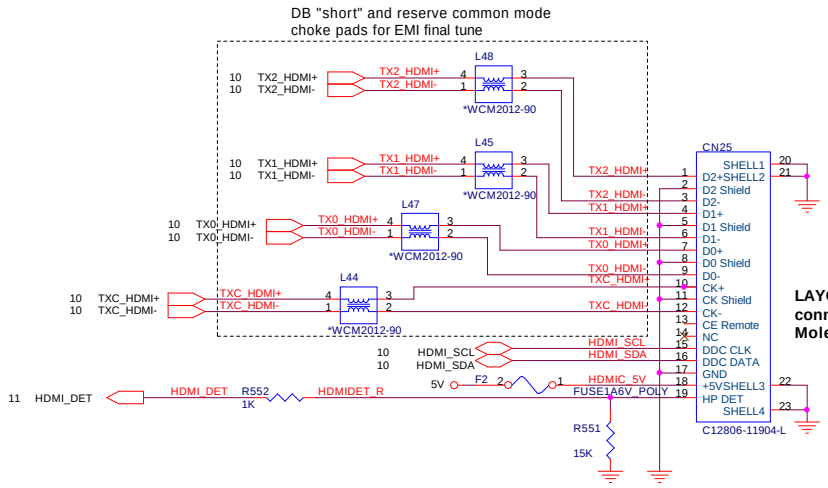
DDCCLK_R
DDCDAT_R
DISPON
BLON
RSTBTN#
MCP51LID#
SIO_PME#
RI#
+1.2V_OV0
+1.2V_OV1
+1.2V_OV2
CR_VID0
CR_VID1
CR_VID2
CVID5
CVID4
CVID3
CVID2
CVID1
CVID0
MCP51TESTMODE
MIIRST#
MCP_TV_EN
SCI#
BOARDID0
BOARDID1
BOARDID2
PCLK_SMB
PDAT_SMB
SMB_ALERT#
CGCLK_SMB
CGDAT_SMB
DDCCLK_R
HT_VLD
HT_VLD_5V#
HT_VLD_RC
HTVDD_EN
HTVDD_EN
HTVDD_EN1
HTVDD_EN1_15V
DDCCLK_R1
EDIDCLK_R1
DDCCLK_R
EDIDCLK_R
DDCCLK_R1
DDCDAT_R
EDIDCLK_R1
EDIDDAT_R
GATEA20
RCIN#
ECPWROK
DNBSWON#
RSTBTN#
KBSMI#
MCP51LID#
PCIE_WAKE#
SIO_PME#
RI#
PM_BATLOW#
HTVDD_EN
HT_VLD
EDIDCLK_R
EDIDDAT_R
SMB_ALERT#
BKPWM
MCP_THERMIP#
SUSBR#
SUSCR#
SUSCK
SLP_DEEP#
PM_BATLOW#
V3P3_DEEP
BOARDID2
BOARDID1
BOARDID0
MCP51_GPIO5
GPIO47
1.2V
1.2V_DUAL
MCP_1.2VHT
1.5V_PLL_USB
1.5V_PLL_SP_VDD
3V_PLL_SP_SS
MCP51_NC1
MCP51_NC2
MCP51_3.3V
MCP51_3.3V_HT
+3.3V_ALW_MCP
+3.3V_USB_DUAL
MCP51_1.5V_SP
1.5V
+1.2V_PLL_MAC_DUAL
+3V_PLL_MAC_DUAL
1.5V_PLL_LEG
3V_PLL_SP_SS
3V_PLL_USB
1.5V_PLL_SP_SS
MCP51_NC5
MCP51_NC4
MCP51_NC3
3V_PLL_USB
MCP51_3.3V_HT
+3V_PLL_MAC_DUAL
+1.2V_PLL_MAC_DUAL
+1.2V_PLL_MAC_DUAL
+1.2V_PLL_MAC_DUAL
+3V_PLL_MAC_DUAL
+3V_PLL_MAC_DUAL
3V_S5



HDMI PORT



LAYOUT must support connectors from JAE, Molex and Acon.

NB5/RD2/HW1	PROJECT : AT8 Quanta Computer Inc.	
	Size: Custom Document Number: LCD CONN,HDMI CONN	Rev: 1A
	Date: Thursday, December 29, 2005 Sheet 18 of 38	

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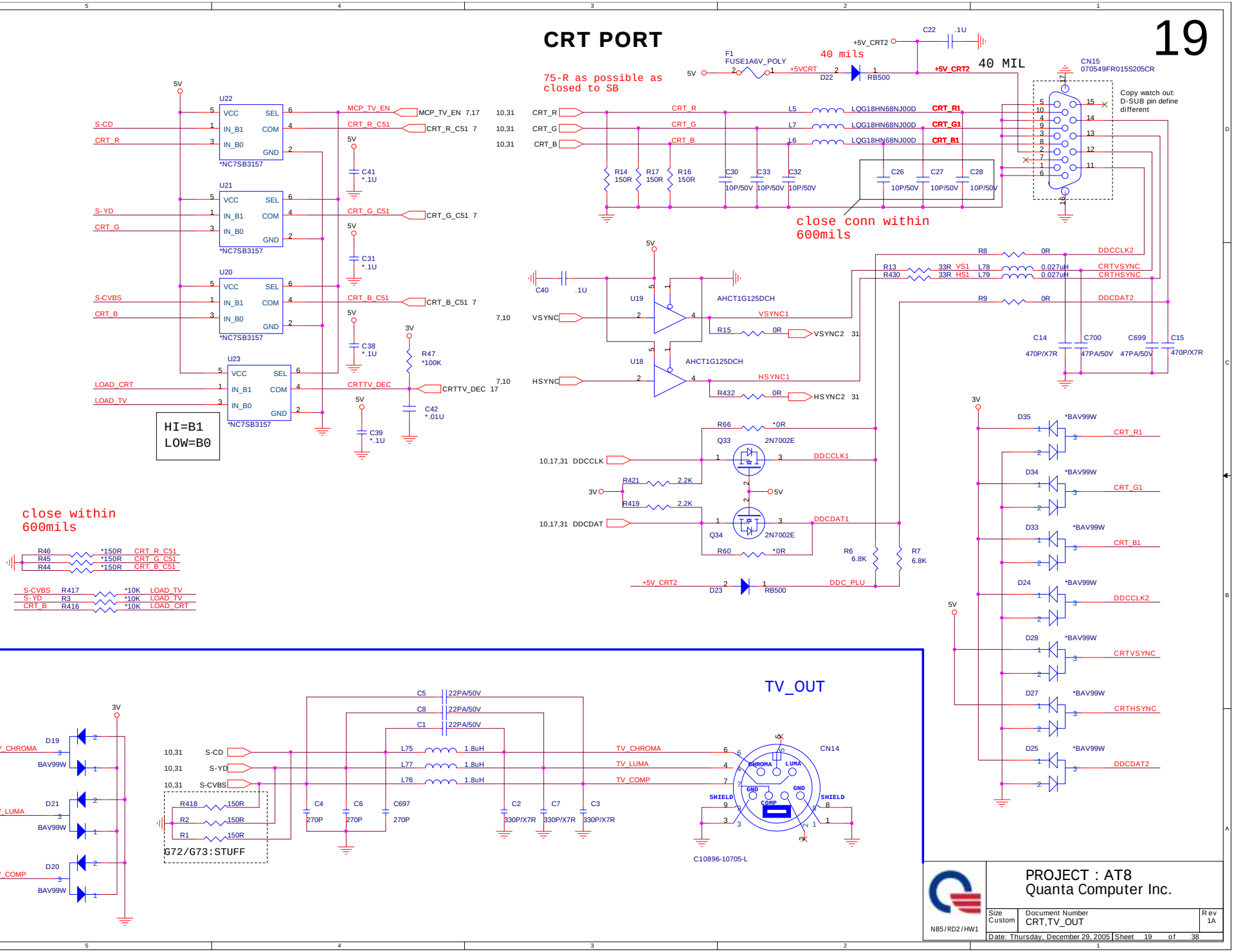
CRT PORT



PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	CRT,TV_OUT	1A

NB5/RD2/HW1

Date: Thursday, December 29, 2005 | Sheet 19 of 38

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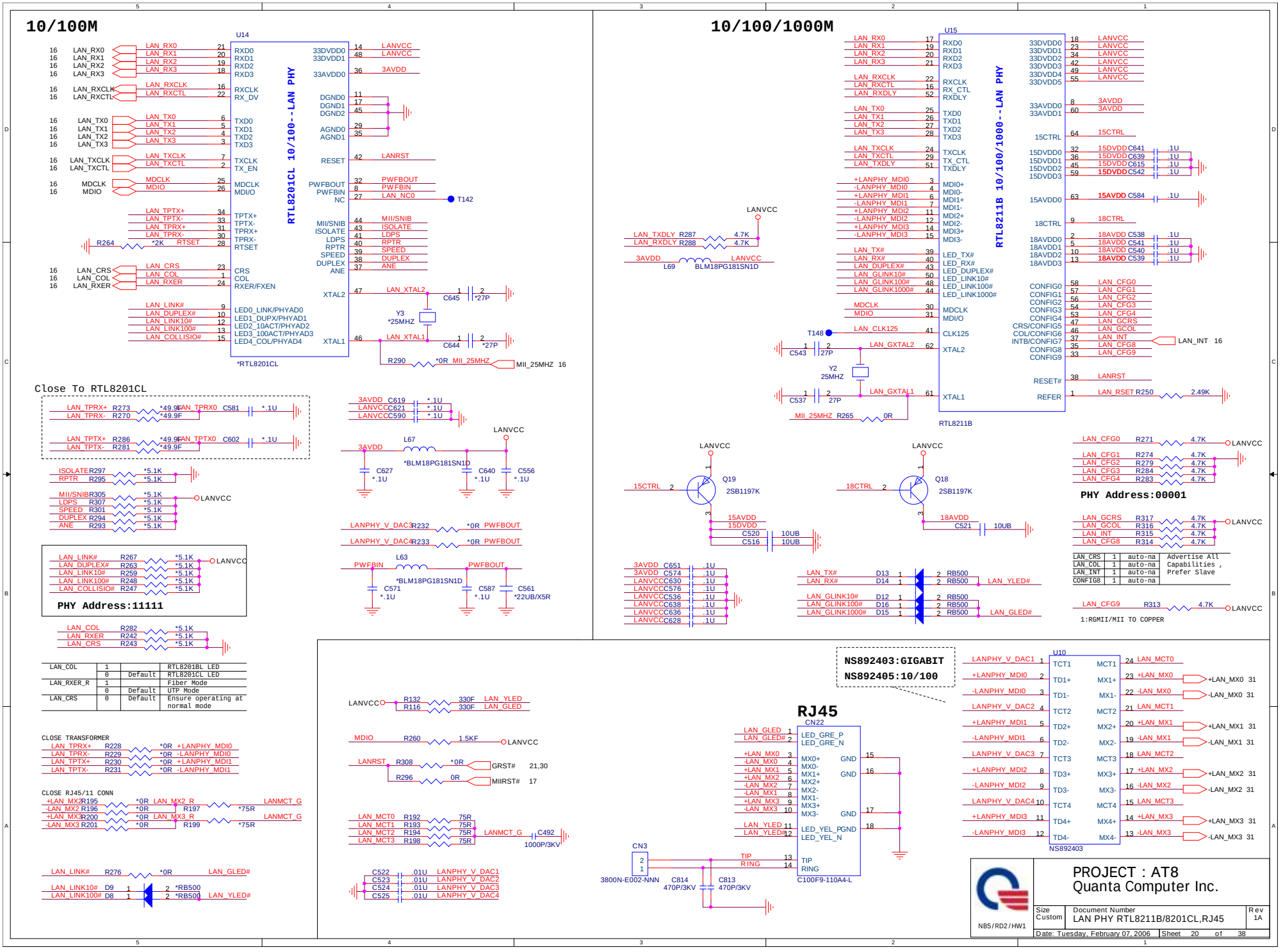

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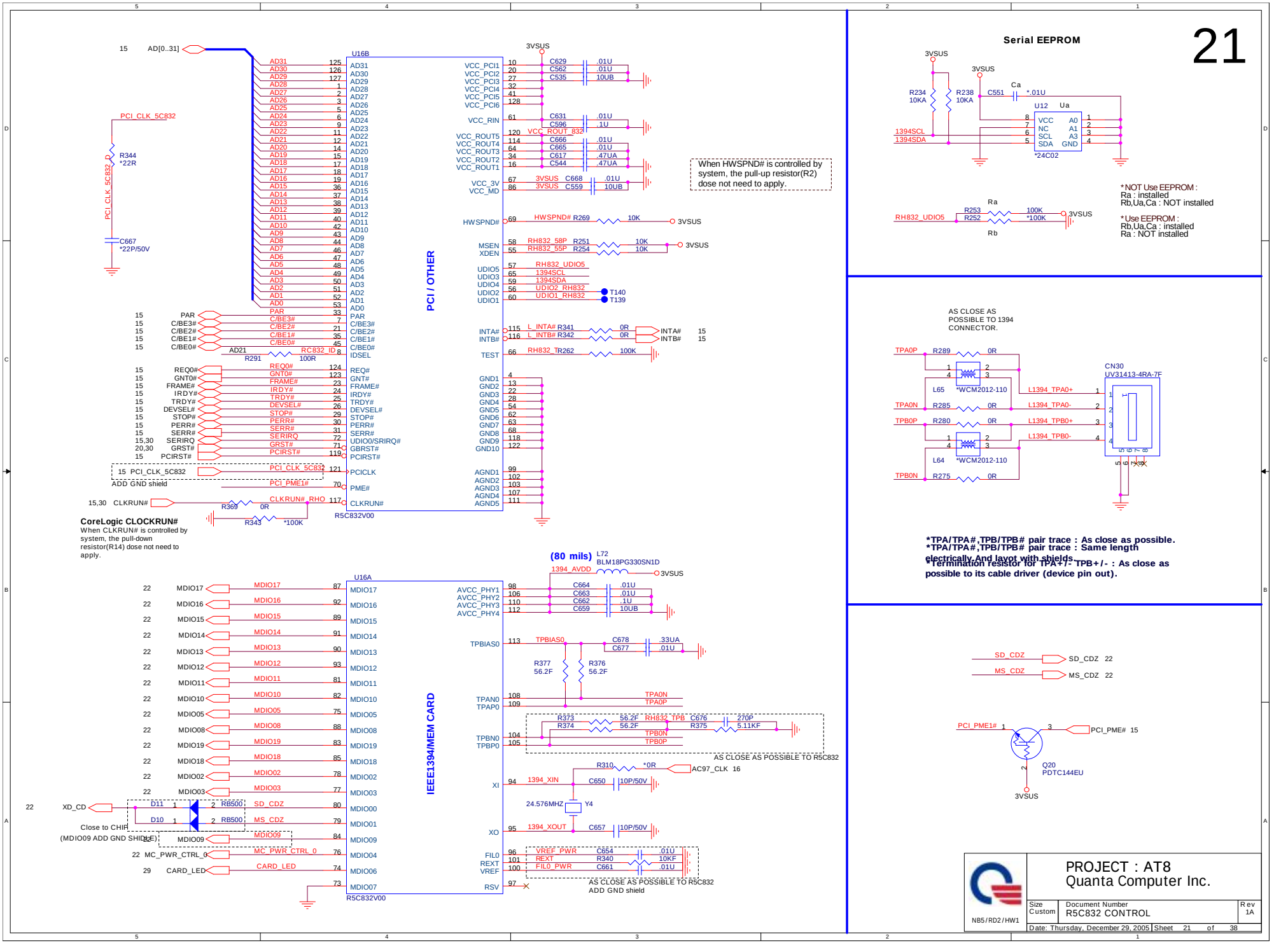

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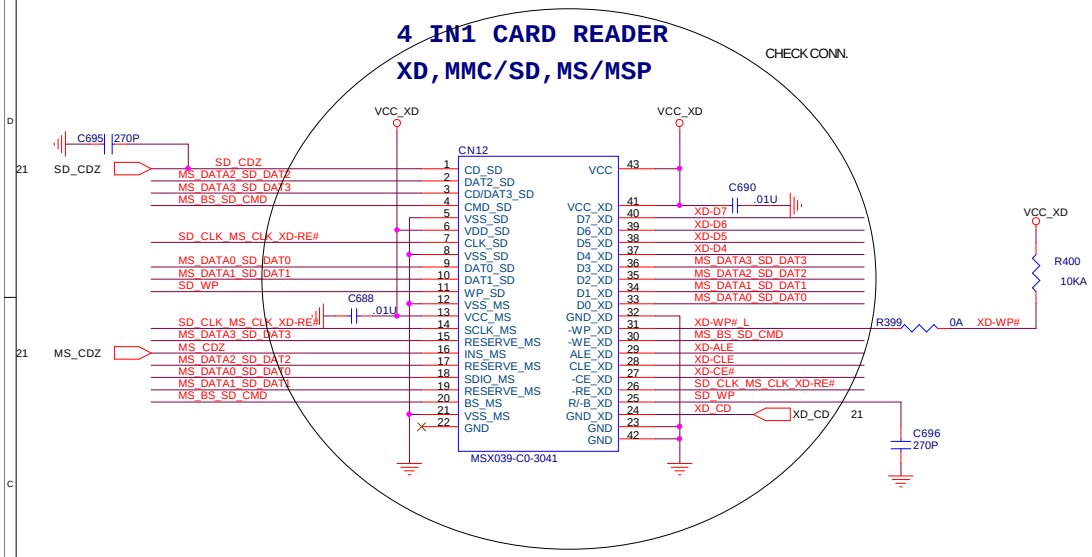
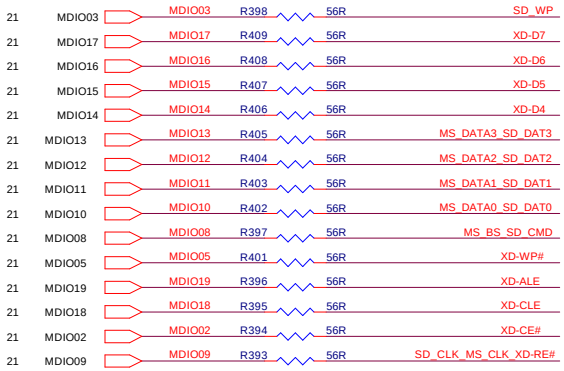
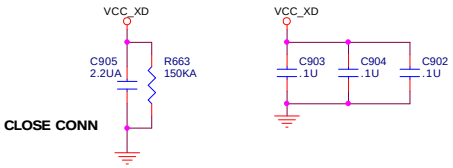
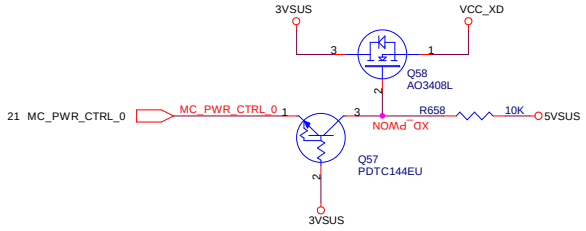
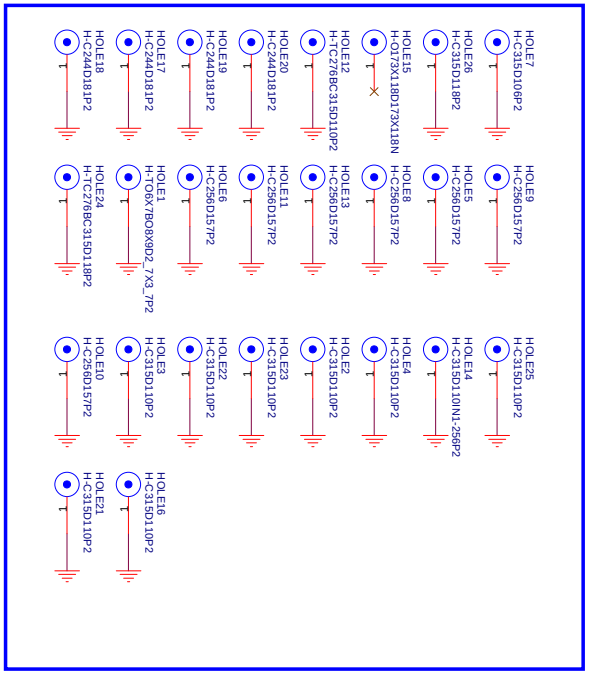




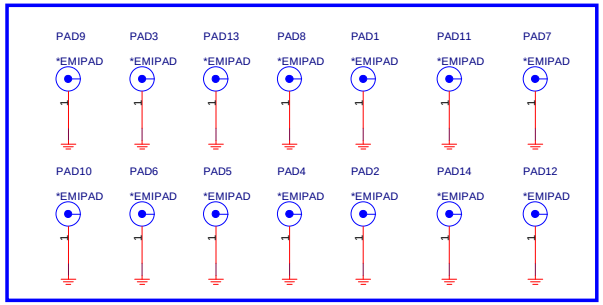




SCREW HOLE



EMI PAD



PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	CARD READER CONN	1A

NB5/RD2/HW1

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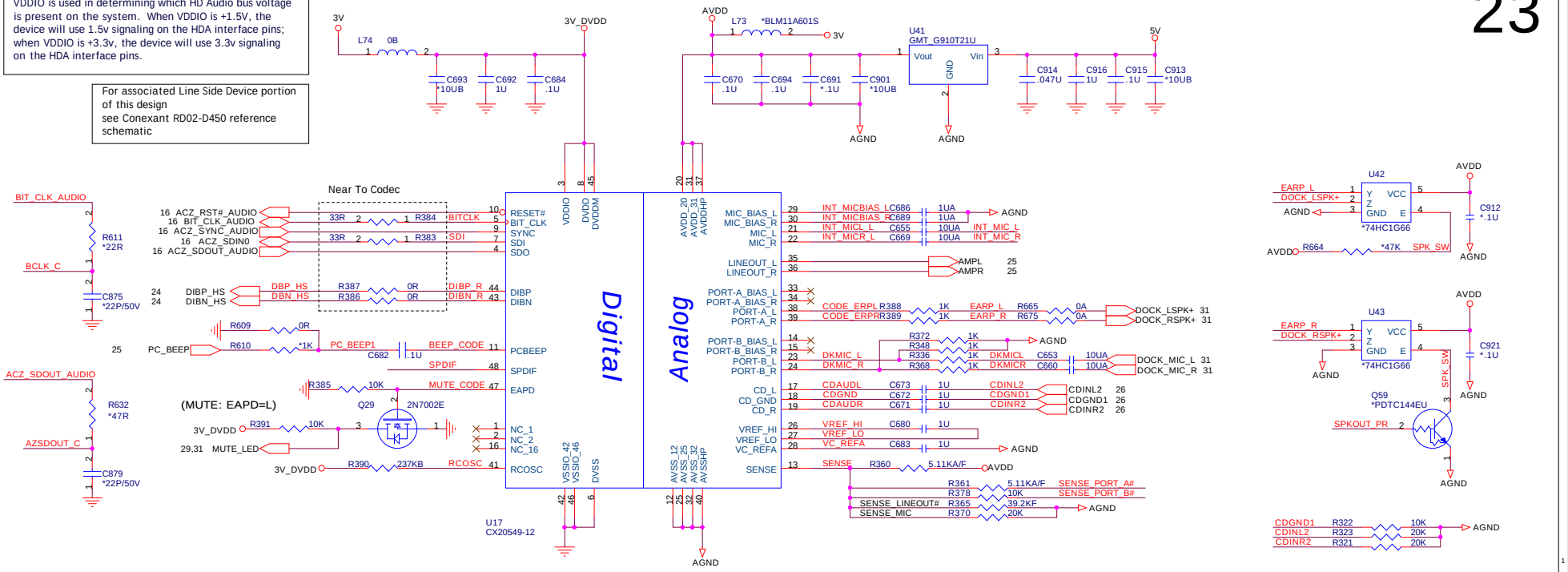
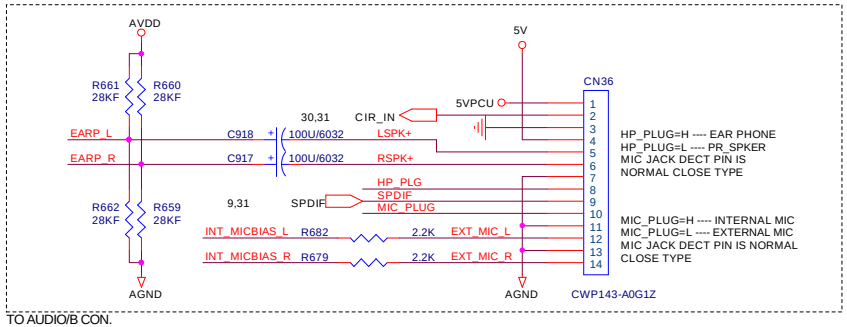
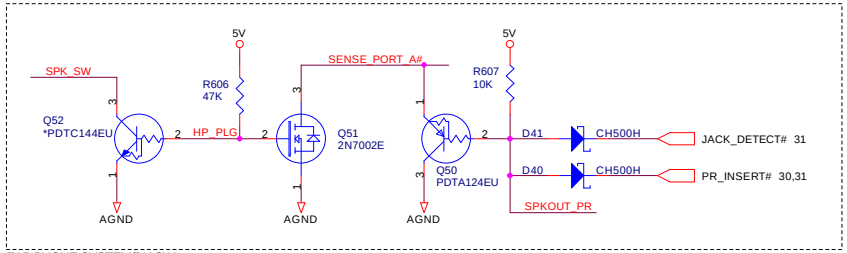
VDDIO is used in determining which HD Audio bus voltage is present on the system. When VDDIO is +1.5V, the device will use 1.5v signaling on the HDA interface pins; when VDDIO is +3.3v, the device will use 3.3v signaling on the HDA interface pins.

For associated Line Side Device portion of this design see Conexant RD02-D450 reference schematic

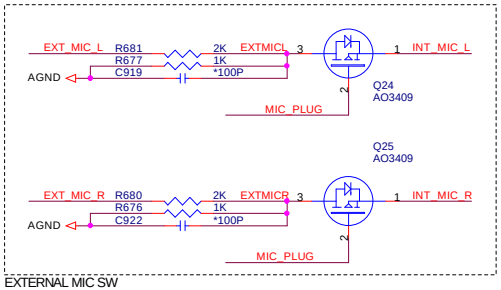




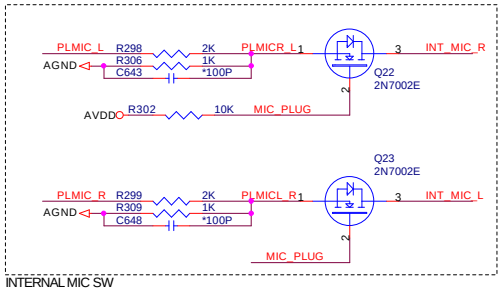
TO AUDIO/B CON.



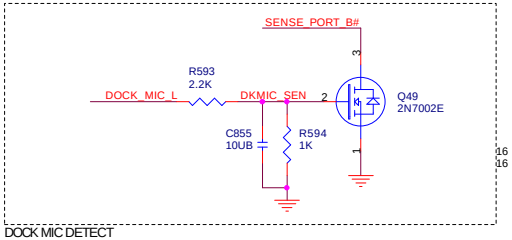
EAR PHONE SYSTEM/DK SW



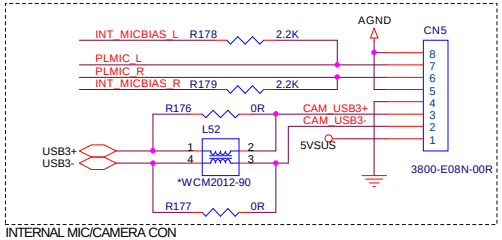
EXTERNAL MIC SW



INTERNAL MIC SW



DOCK MIC DETECT



INTERNAL MIC/CAMERA CON

NB5/RD2/HW1	PROJECT : AT8 Quanta Computer Inc.	
Size: Custom	Document Number: Azalia CTRL_CONEXANT20549-12	Rev: 1A
Date: Thursday, December 29, 2005	Sheet 23 of 38	

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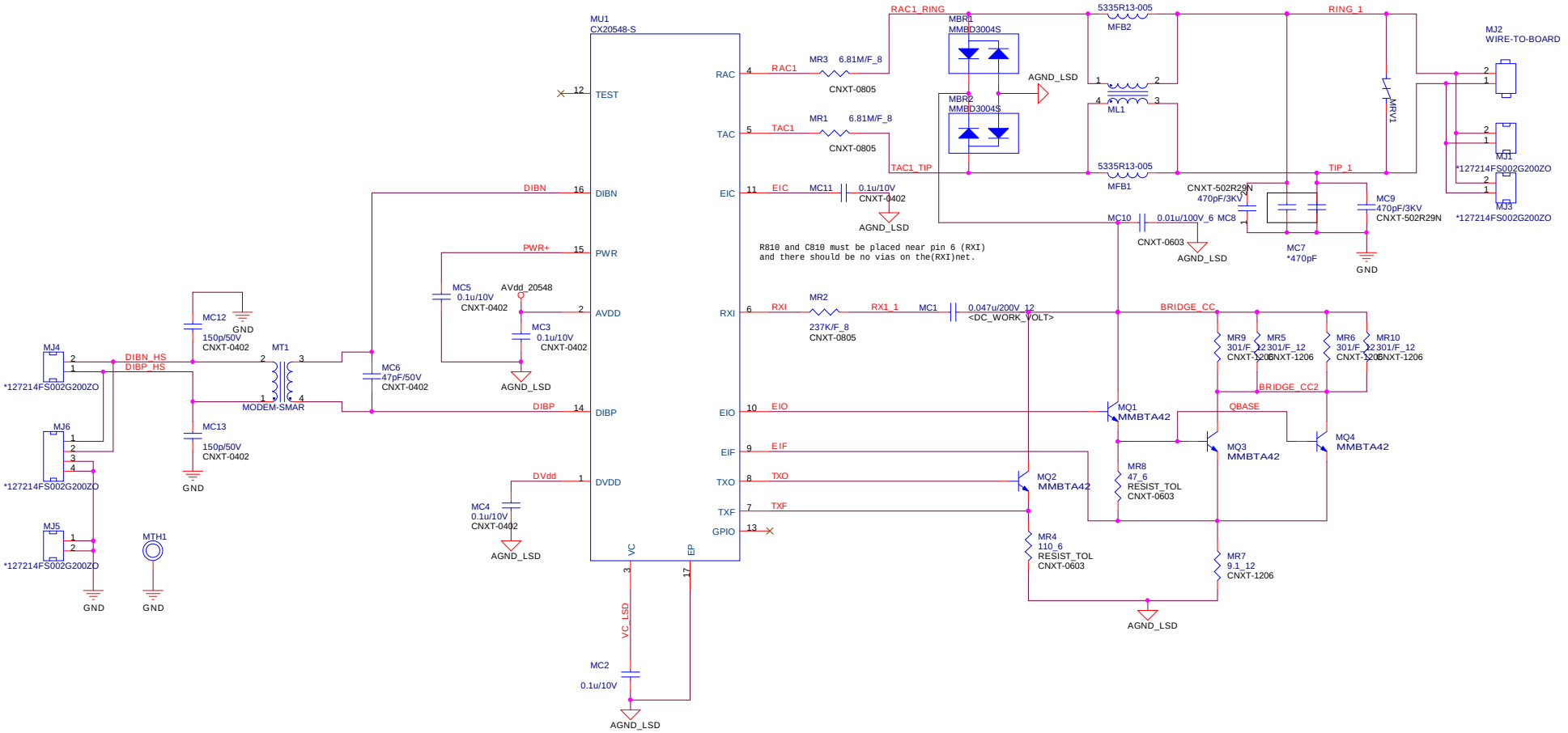
Revision History		
REV	Description	Date
0	Initial Release	April 26, 2005

23 DIBN_HS
23 DIBP_HS



NB5/RD2/HW1	PROJECT : AT8 Quanta Computer Inc.		
	Size: Custom	Document Number: MODEM (DAA)	Rev: 1A
	Date: Thursday, December 29, 2005	Sheet 24 of 38	

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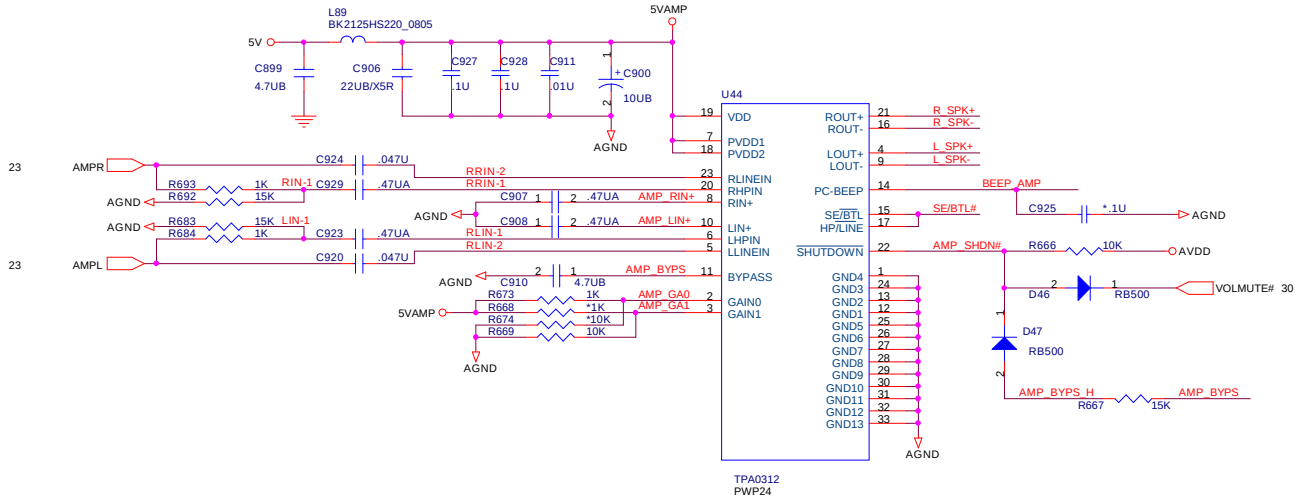
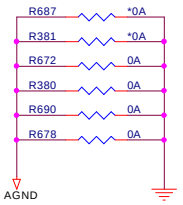
AUDIO AMPLIFIER

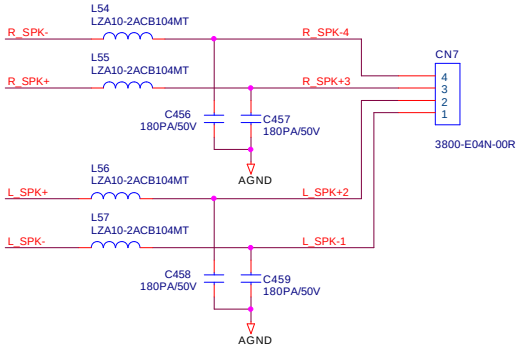


0312 Gain Table

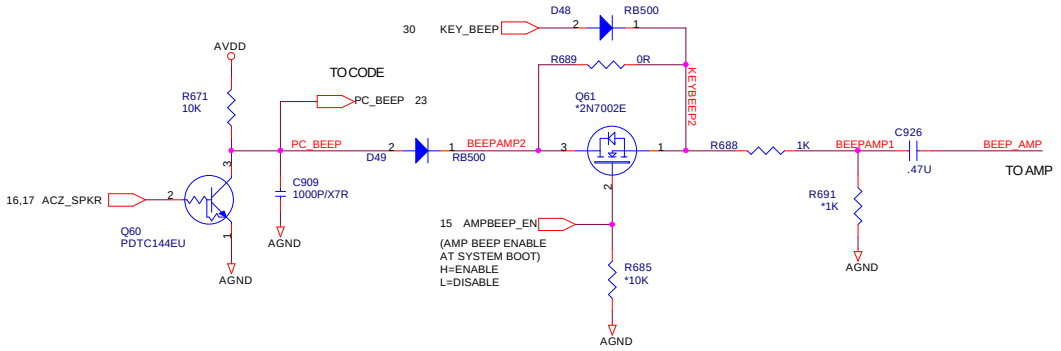
GAIN0	GAIN1	SE/BTL	AV(INV)
0	0	0	6dB
0	1	0	10dB
1	**0**	**0**	**15.6dB**
1	1	0	21.6dB
x	x	1	4.1dB



INT. SPEAKER



PCSPK BEEP



NB5/RD2/HW1	PROJECT : AT8 Quanta Computer Inc.		
	Size: Custom	Document Number: JACK, AMP_TPA0312	Rev: 1A
	Date: Thursday, December 29, 2005	Sheet 25 of 38	

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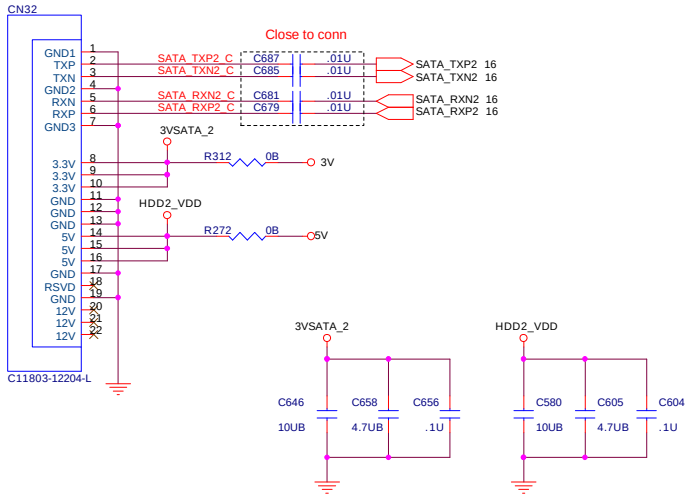
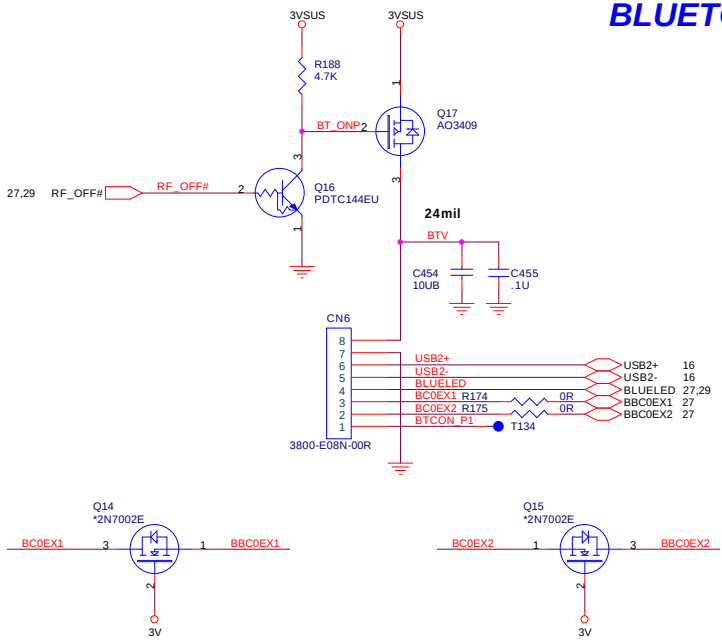
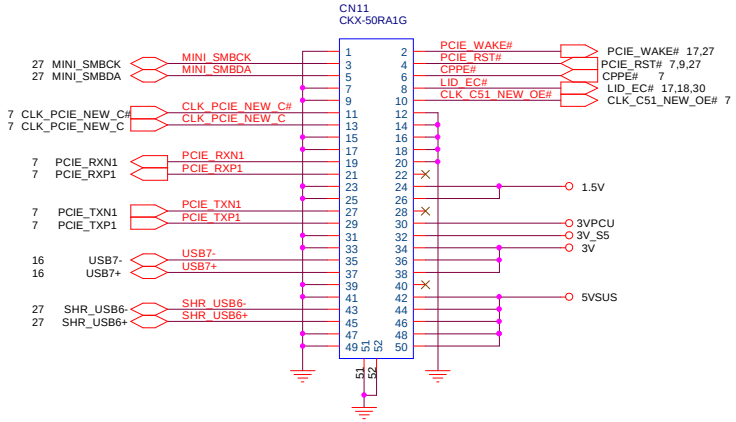
SATA_2 CONNECTOR
For 17"W Second HDD

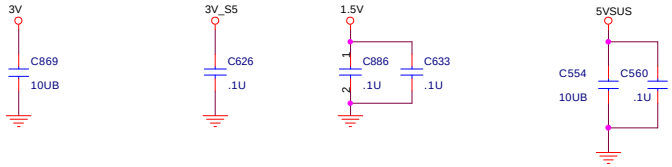


BLUETOOTH



NEWCARD





NB5/RD2/HW1	PROJECT : AT8 Quanta Computer Inc.	
Size: Custom	Document Number: NEW CARD/BT	Rev: 1A
Date: Thursday, December 29, 2005	Sheet 28 of 38	

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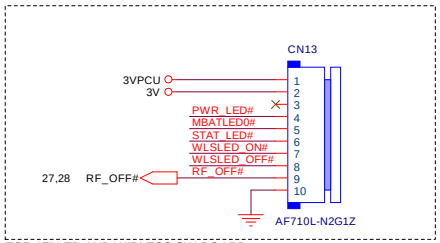
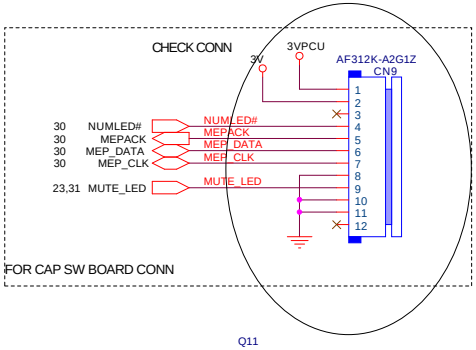
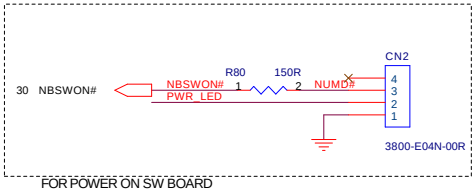


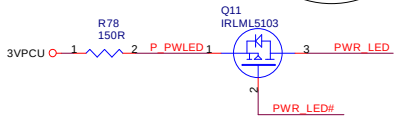
FOR 17" LED AND WIRLESS SW BOARD

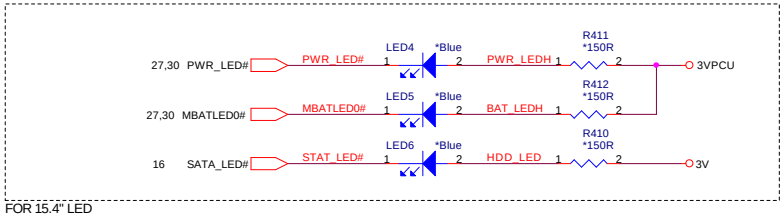


FOR CAP SW BOARD CONN

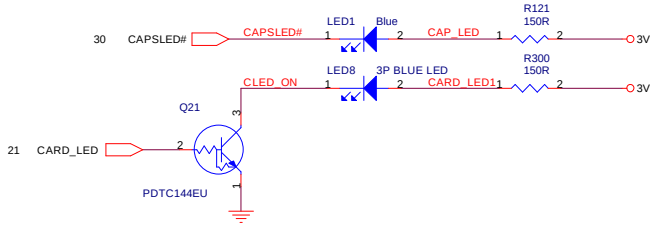


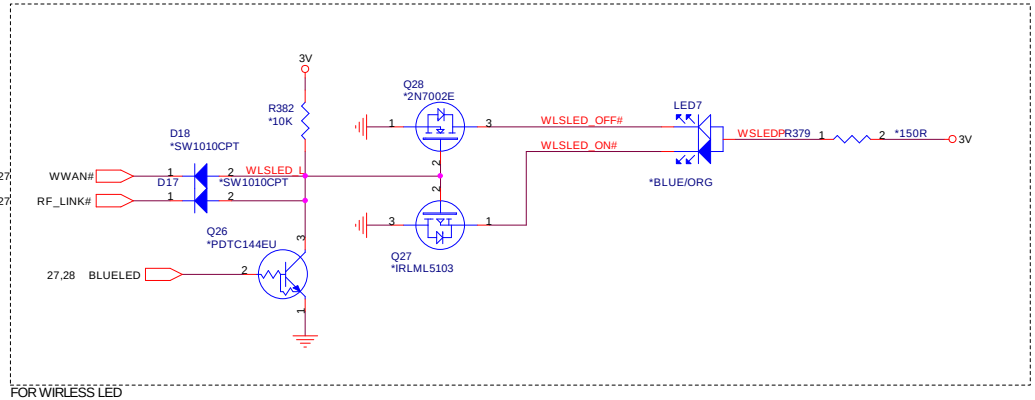
FOR POWER ON SW BOARD



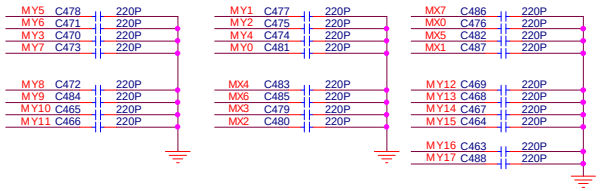


FOR 15.4" LED

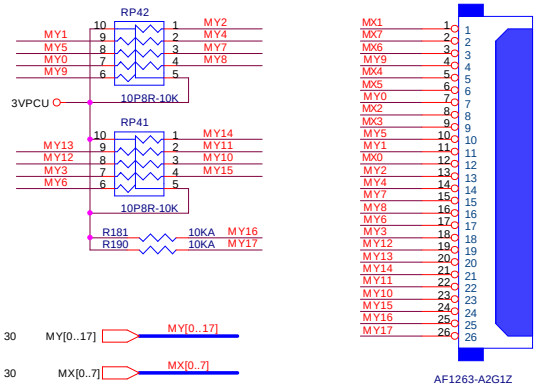


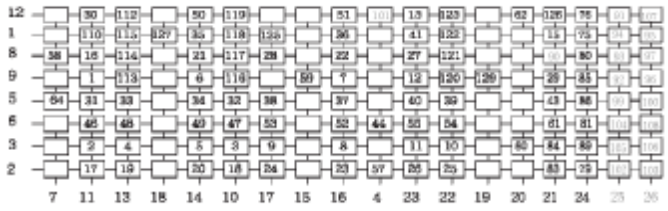


FOR WIRLESS LED



KEYBOARD PULL-UP





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8	38	16	114		33	117	28		22		27	121				93	95	97	98				
9		1	113		6	116		50	7	12	120	126		26	85	92	96						
5	64	31	33		34	32	38		37		40	39		43	86		103						
6		45	48		49	47	53		52	44	55	54		61	81		104	108					
3		2	4		5	3	9		8		11	10		80	84	89	105	106					
2		17	19		20	18	24		23	57	56	25		83	75		107	109					
	7	11	13	15	14	10	17	15	16	4	23	22	19	20	21	24	25	26					

NB5/RD2/HW1

PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	LED/KEYBOARD/TP	1A

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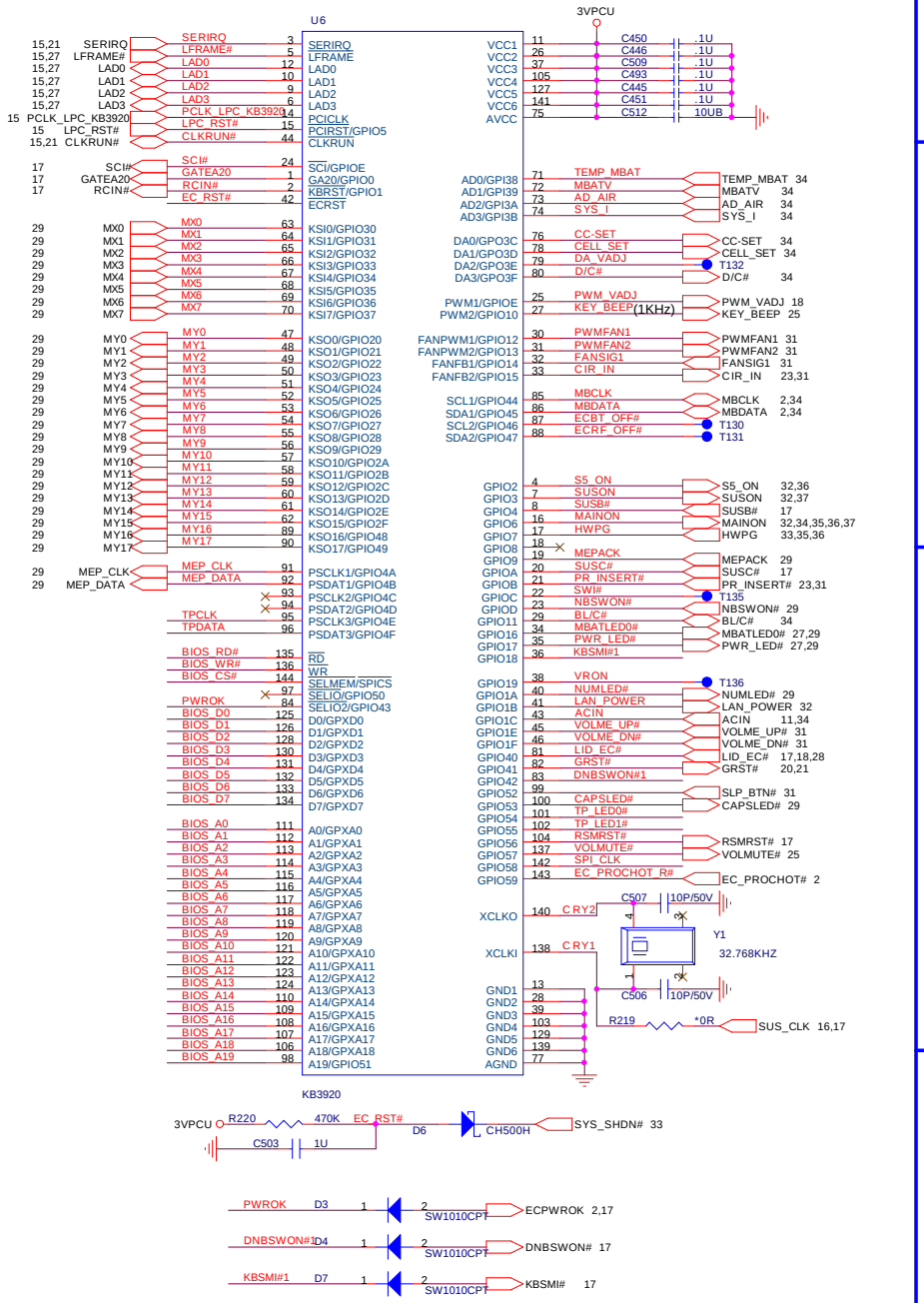
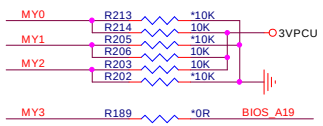
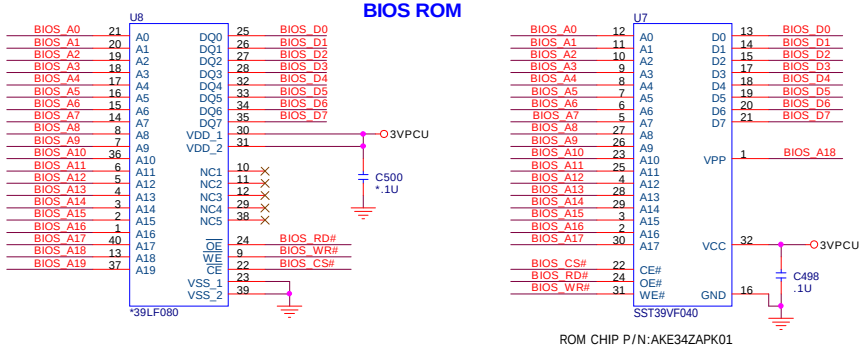


STRAP PIN

MY0	47	TP_TEST: Clock Test Mode Low: Test Mode. HIGH: 32kHz clock in normal runing
MY1	48	TP_PLL: DPLL Test Mode Low: Test Mode. HIGH: Normal operation
MY2	49	TP_SPI: Default flash access Low: Boot from SPI flash part HIGH: Boot from ISA flash part
MY3	50	TP_ISP: In System Programming Mode Low: ISP mode HIGH: Normal Mode

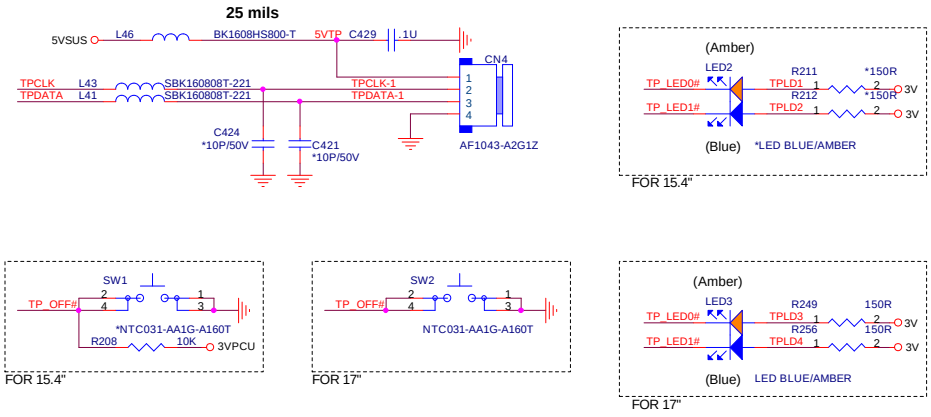


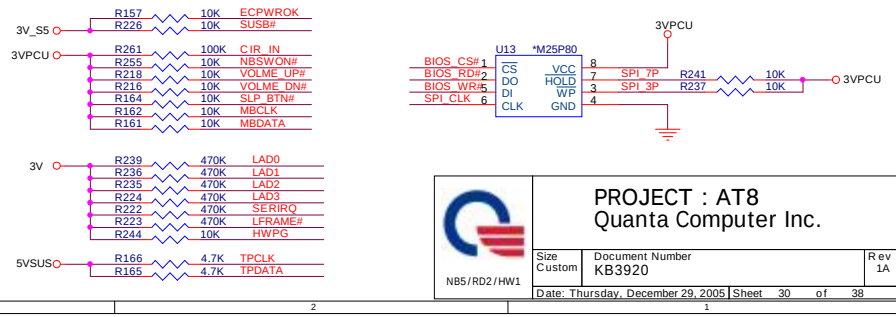
BIOS ROM



ROM CHIP P/N:AKE34ZAPK01

TOUCH PAD CONNECTOR





NB5/RD2/HW1	PROJECT : AT8 Quanta Computer Inc.		
	Size: Custom	Document Number: KB3920	Rev: 1A
	Date: Thursday, December 29, 2005	Sheet 30 of 38	

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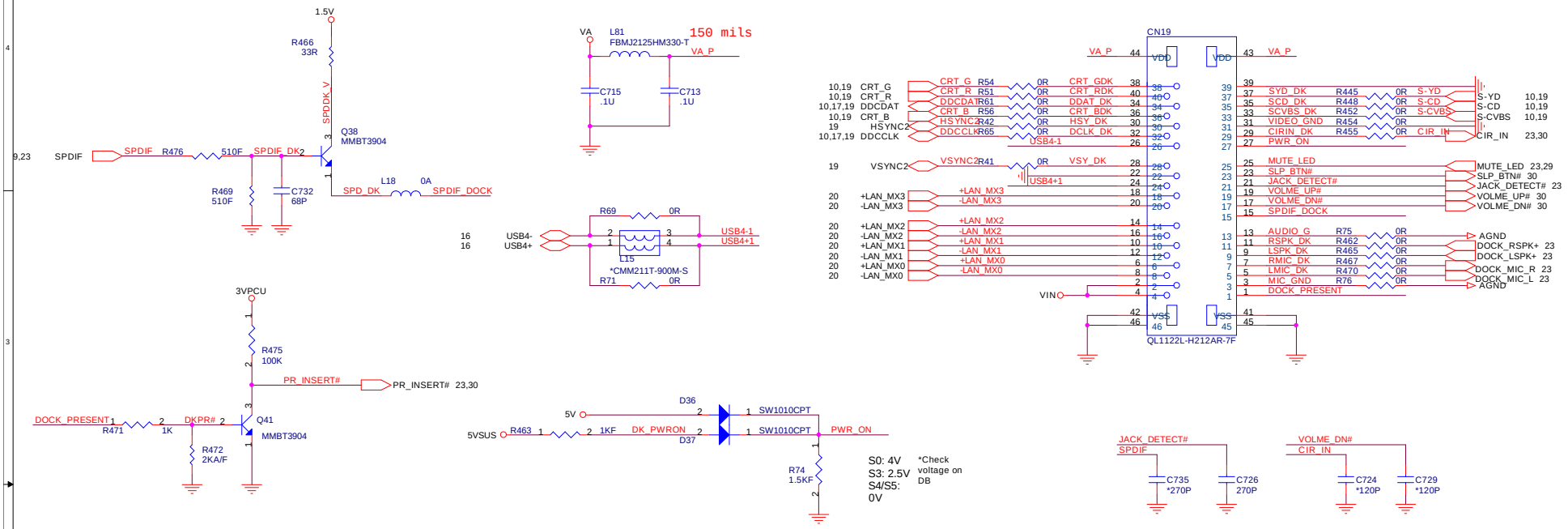
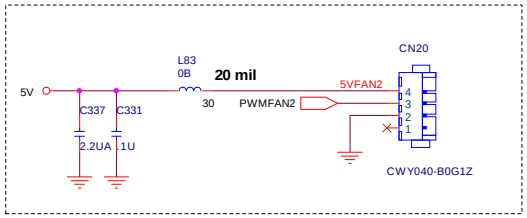
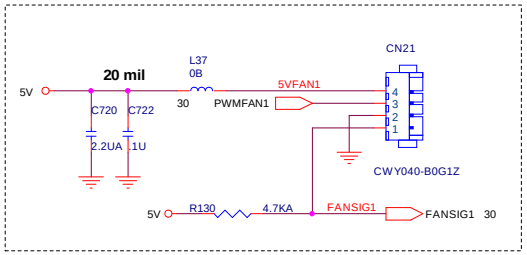
CABLE DOCK



FAN



FAN2 PWM CONNECTOR



FAN1 PWM CONNECTOR

PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	CABLE DOCKING	1A

NB5/RD2/HW1

Date: Thursday, December 29, 2005 Sheet 31 of 38

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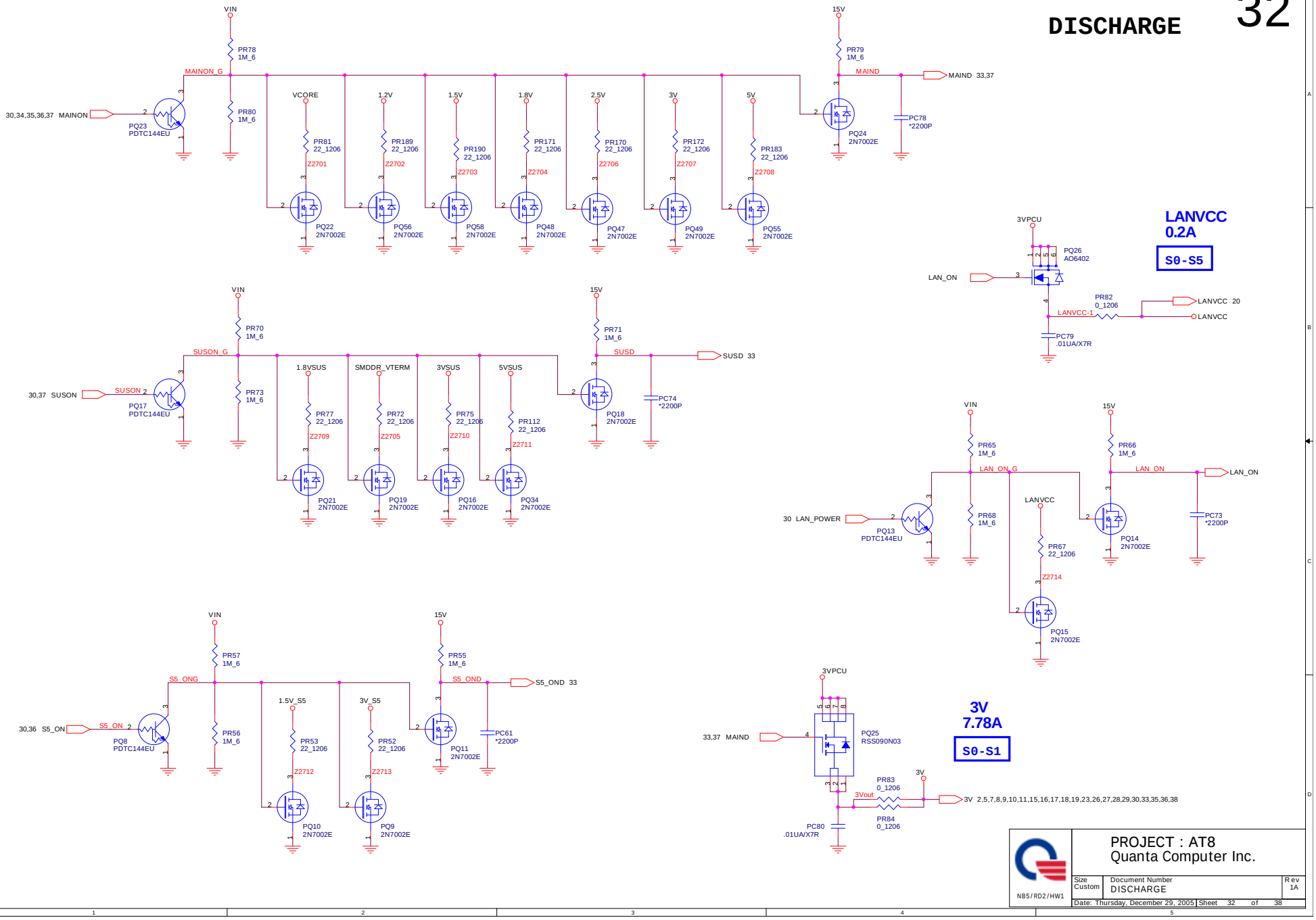
DISCHARGE

32



PROJECT : AT8
Quanta Computer Inc.

Size	Document Number	Rev
Custom	DISCHARGE	1A

NB5/RD2/HW1

Date: Thursday, December 29, 2005 | Sheet 32 of 38

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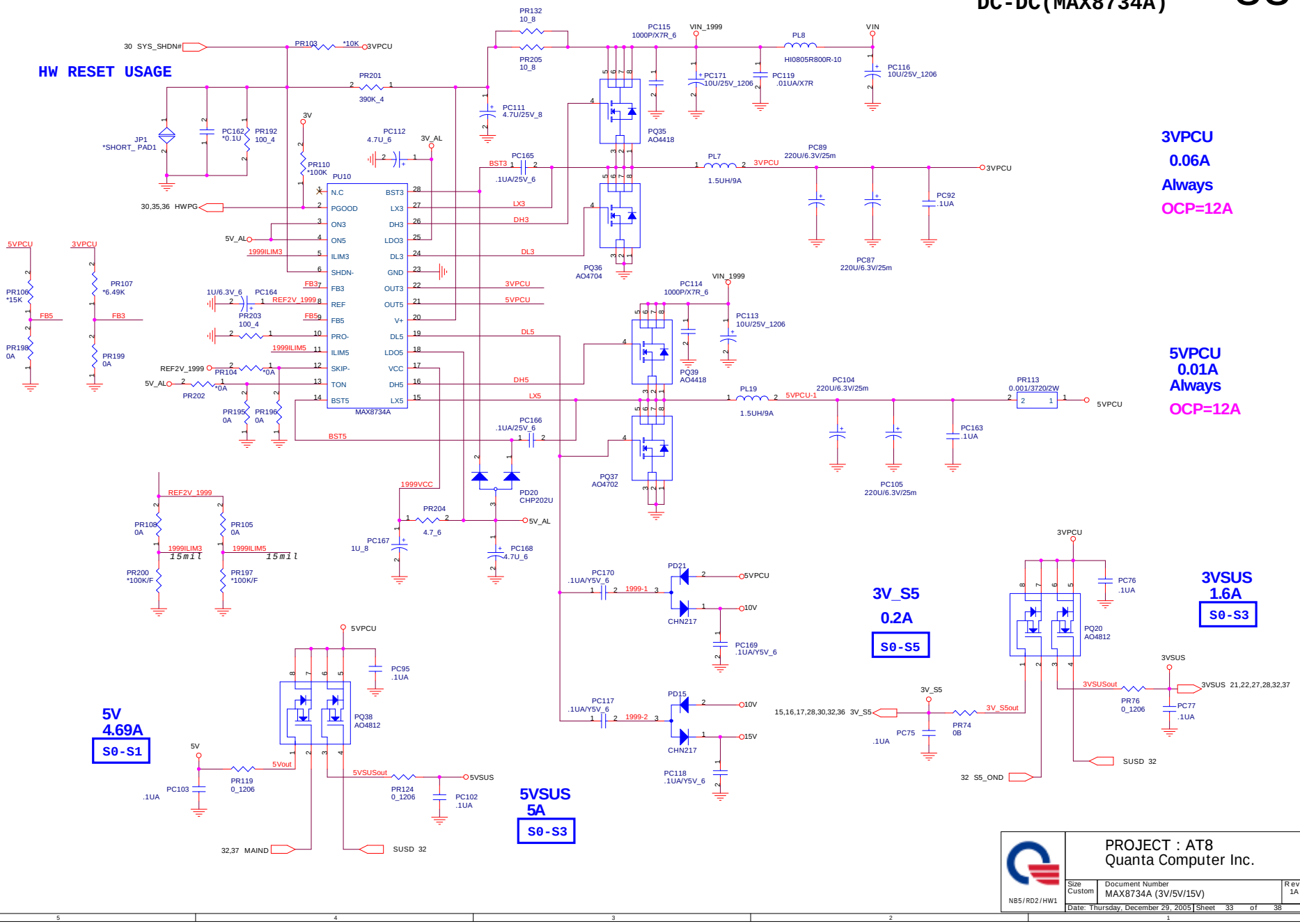

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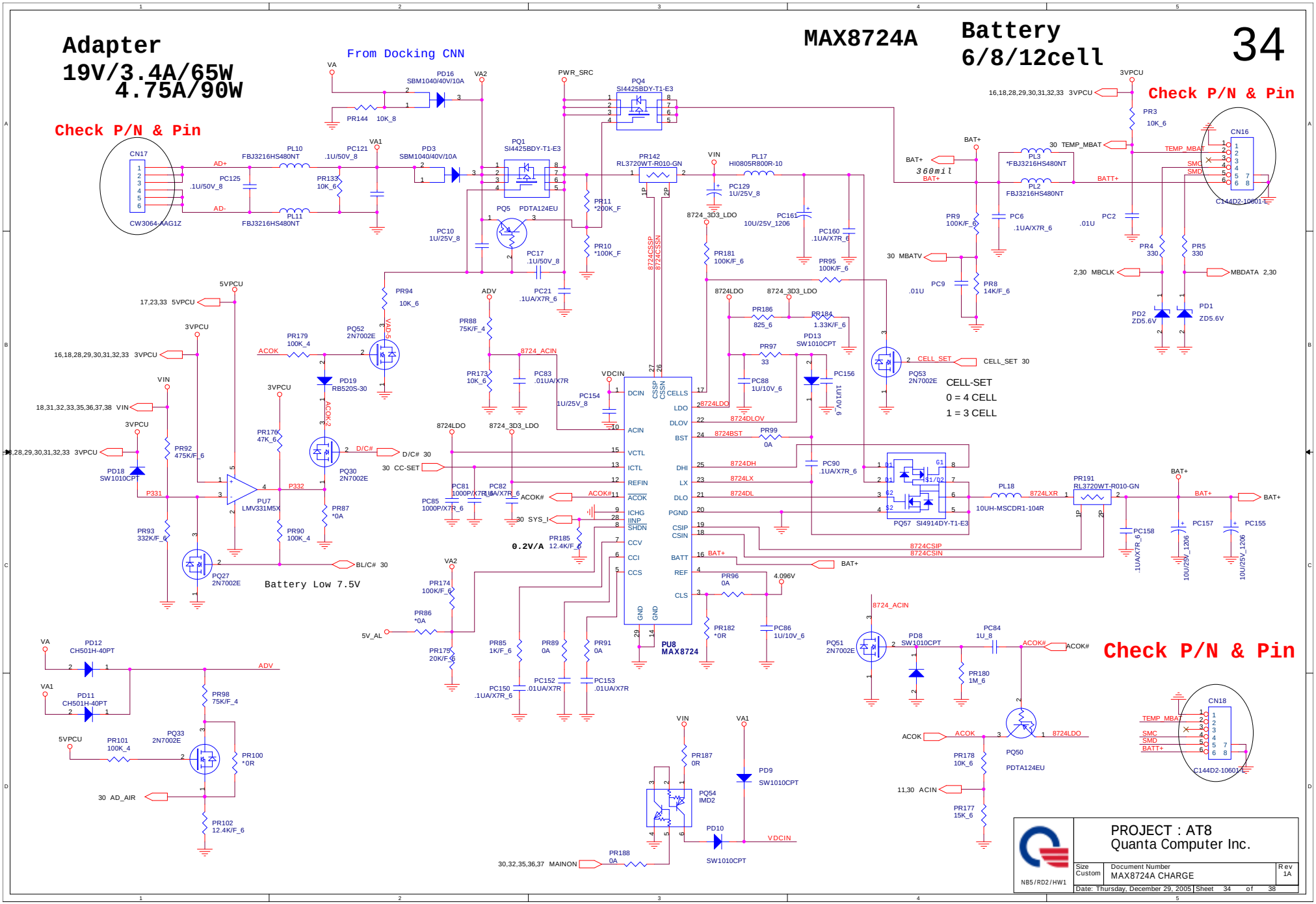

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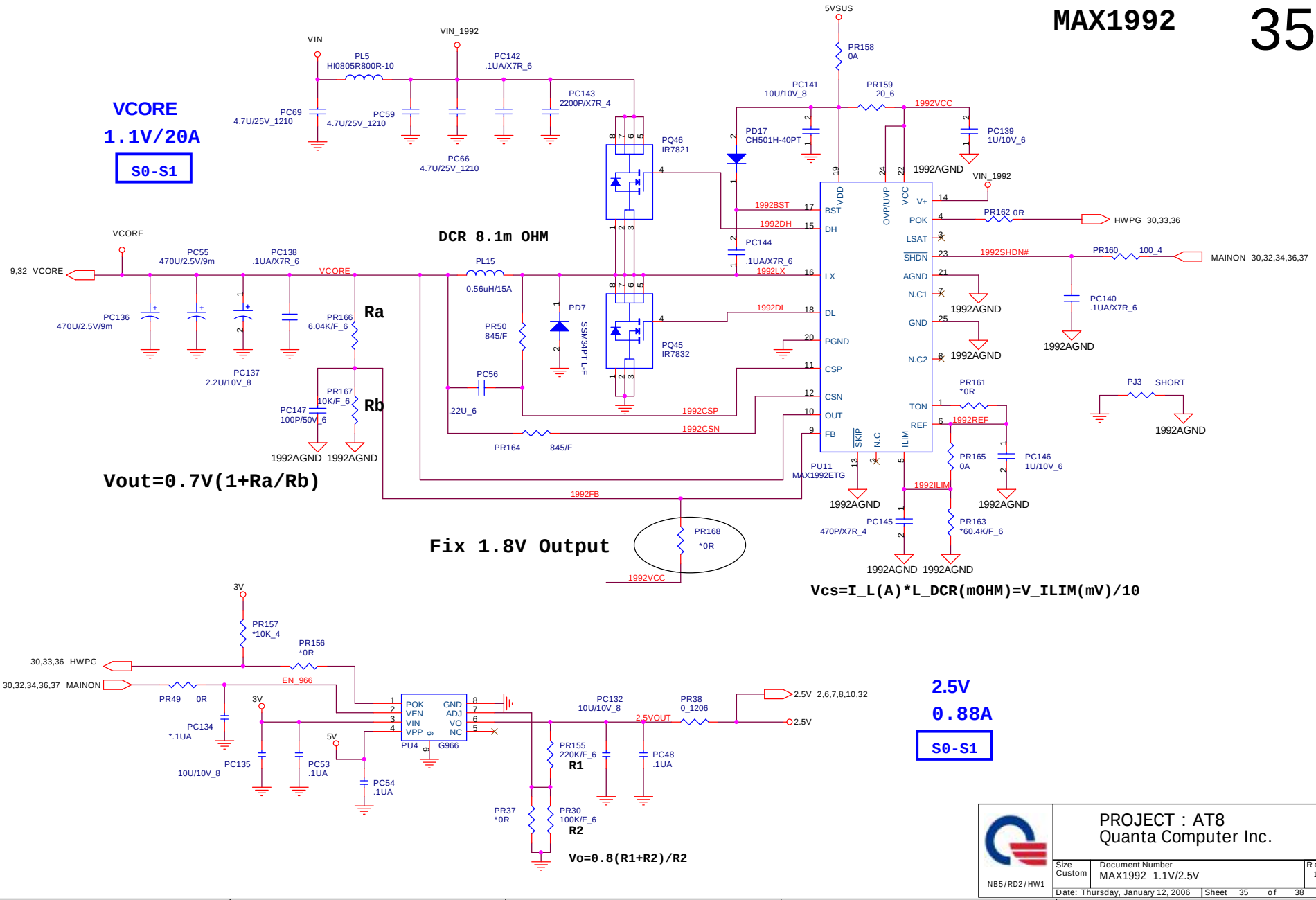

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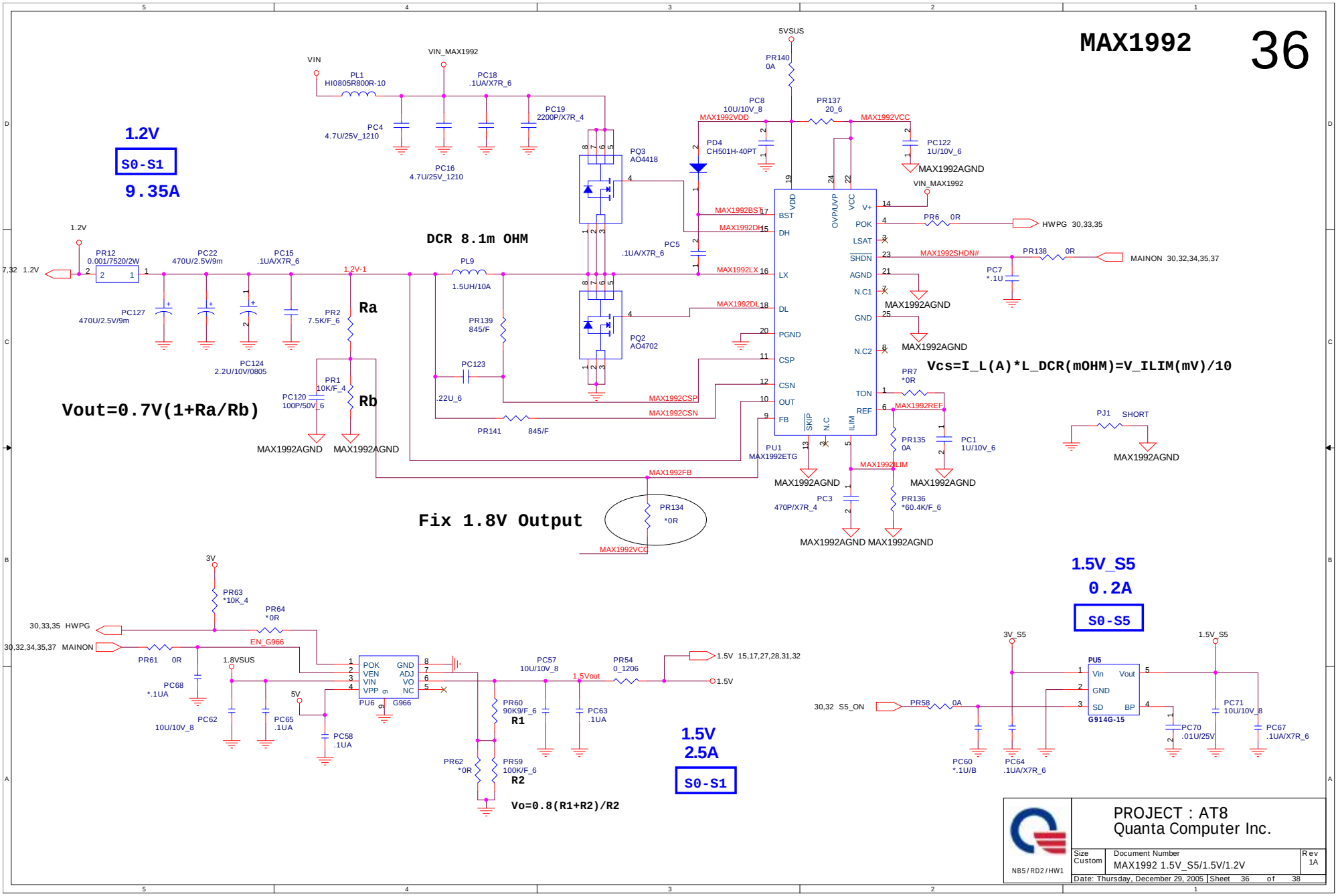

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1.8VSUS 9.1A
S0-S3

1.8 Volt +/-5%
Design Current:15.0A
OCP: Max. 18.0A

DCR 1.1m Ohms+-10%

SMDDR_VTERM 1.53A / 0.9V
S0-S3

0.9 Volt +/-5%
Design Current:1.5A
Maximum Current: 1.8A

SMDDR_VTERM 0.9V ?A
S0-S3

0.9V/1.2A

1.8V 4.75A
S0-S1

PROJECT : AT8
Quanta Computer Inc.

NB5/RD2/HW1

Size	Document Number	Rev
Custom	SC480 1.8VSUS/1.8V/SMDDR	1A

Date: Thursday, December 29, 2005 | Sheet 37 of 38

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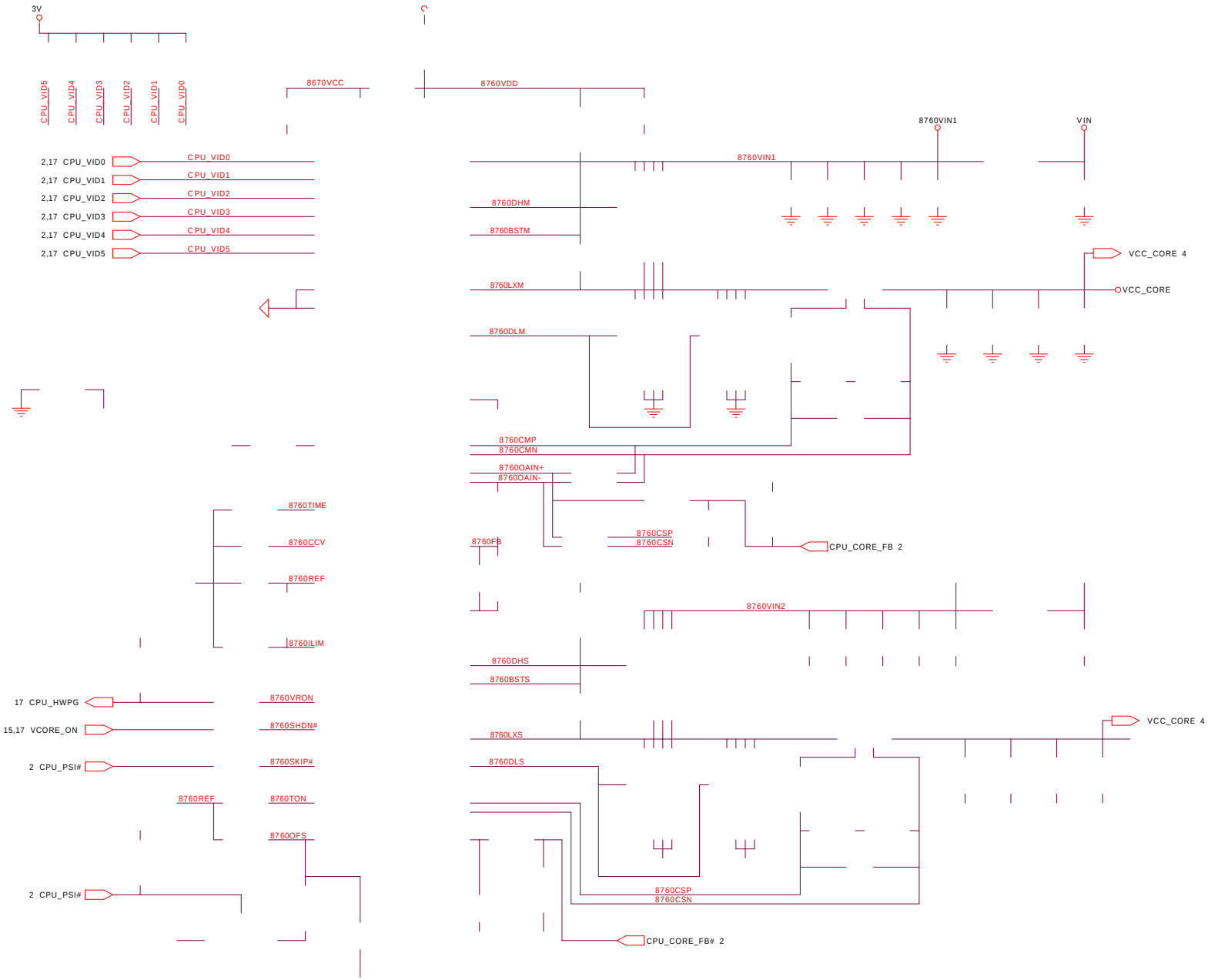
3V
CPU_VID5
CPU_VID4
CPU_VID3
CPU_VID2
CPU_VID1
CPU_VID0
8670VCC
8760VDD
8760VIN1
VIN
2,17 CPU_VID0
2,17 CPU_VID1
2,17 CPU_VID2
2,17 CPU_VID3
2,17 CPU_VID4
2,17 CPU_VID5
8760DHM
8760BSTM
8760LXM
VCC_CORE 4
VCC_CORE
8760DLM
8760CMP
8760CMN
8760OAIN+
8760OAIN-
8760TIME
8760CCV
8760REF
8760ILIM
8760FB
8760CSP
8760CSN
CPU_CORE_FB 2
8760VIN2
8760DHS
8760BSTS
17 CPU_HWPG
15,17 VCORE_ON
2 CPU_PSI#
8760VRON
8760SHDN#
8760SKIP#
8760TON
8760OFS
8760REF
8760LXS
8760DLS
VCC_CORE 4
8760CSP
8760CSN
2 CPU_PSI#
CPU_CORE_FB# 2